

Week 5

Transcriptomics — From RNA Molecules to Biological Discovery

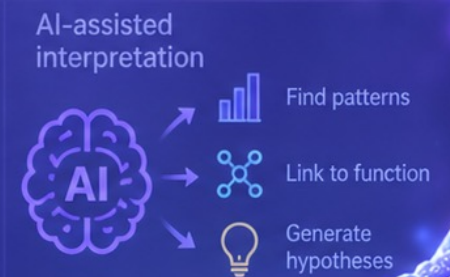
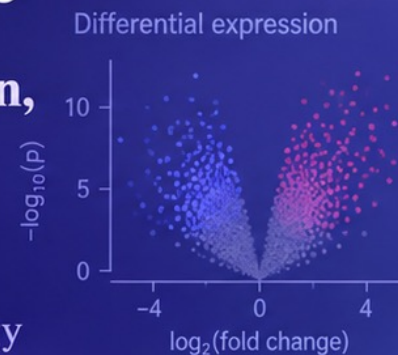
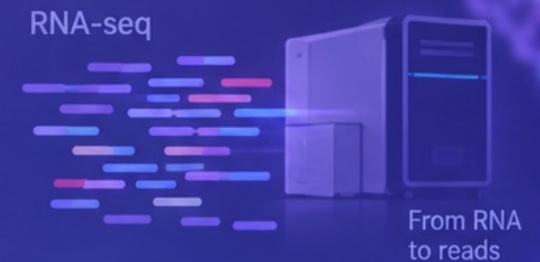
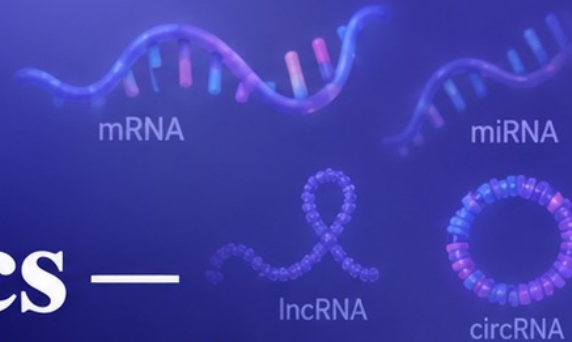
**RNA Classes, RNA-seq, Differential Expression,
and AI-Assisted Interpretation**

Bioinformatics: From Multi-Omics Data to Discovery

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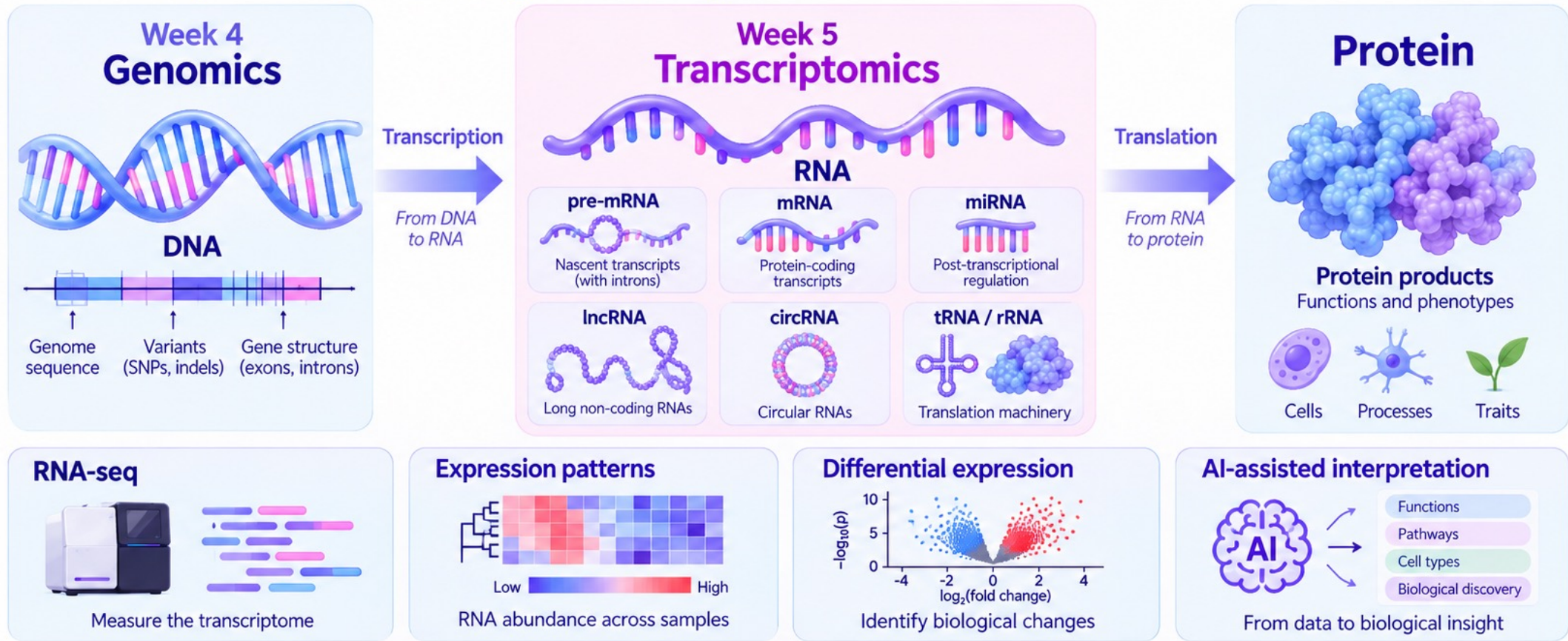
Two 40-Minute Sessions | Fall 2026



Week 5: Transcriptomics

From central dogma to the RNA layer: connecting genomics, RNA classes, RNA-seq, and biological discovery

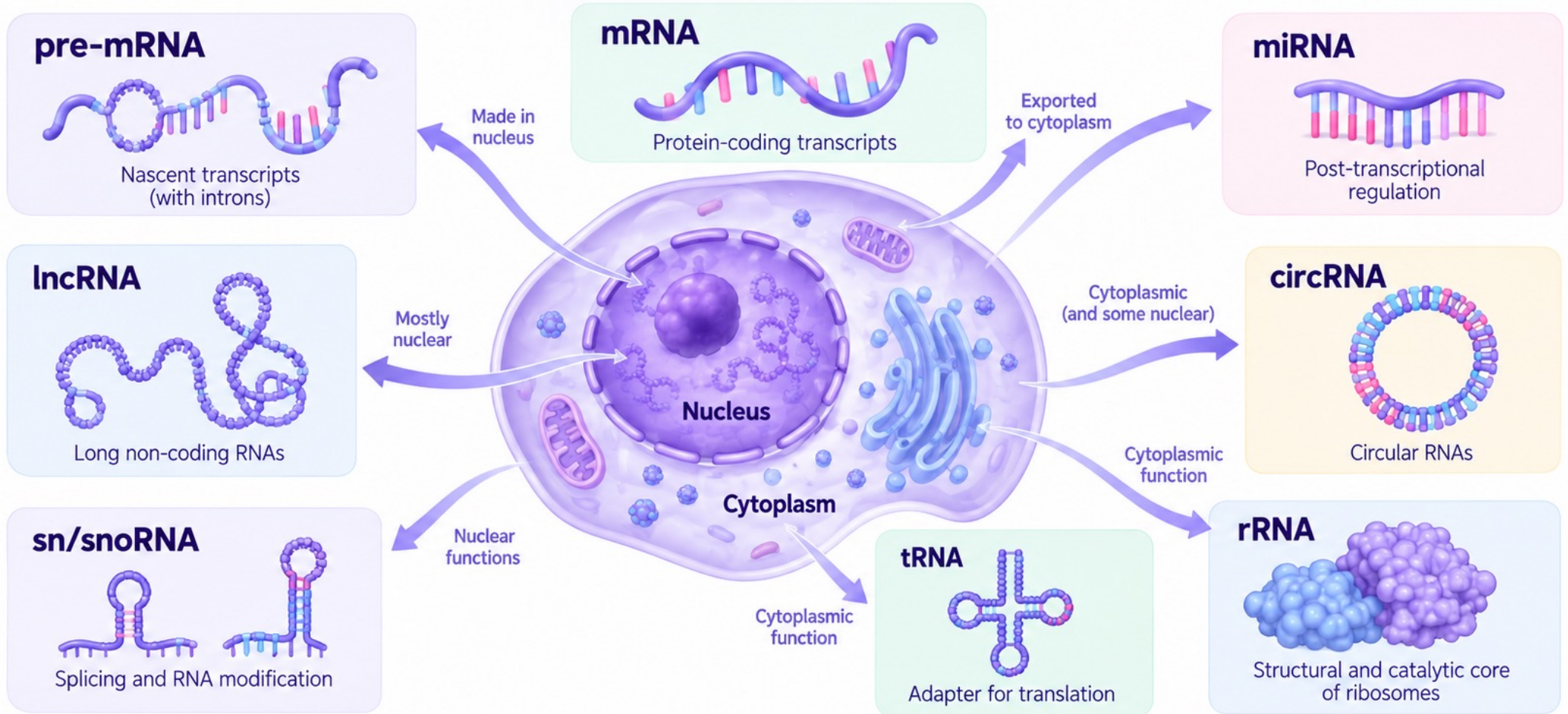
Week 5 | Slide 1



Transcriptomics connects genome information to RNA function, expression, and discovery.

One biological sample contains many RNA worlds

Week 5 | Slide 2

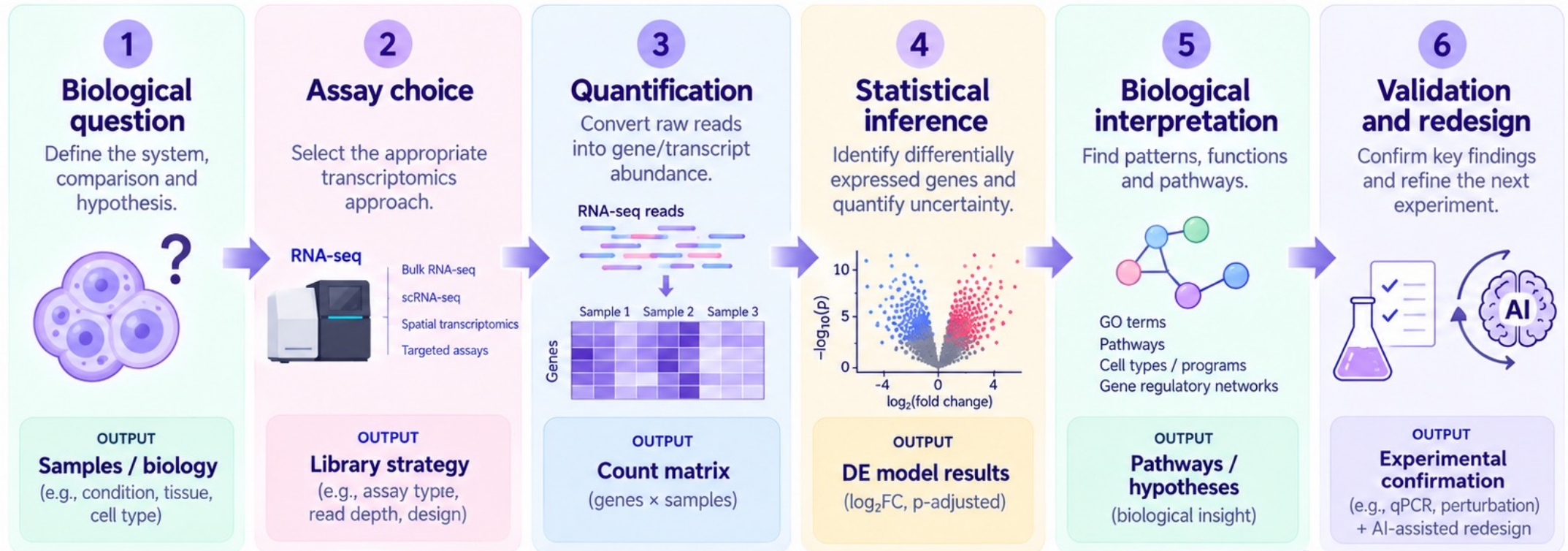


A single sample contains **many RNA species** with different structures, functions, and measurable signals.

The evidence chain of transcriptomics

Week 5 | Slide 3

From a biological question to validated insight, each step builds evidence.

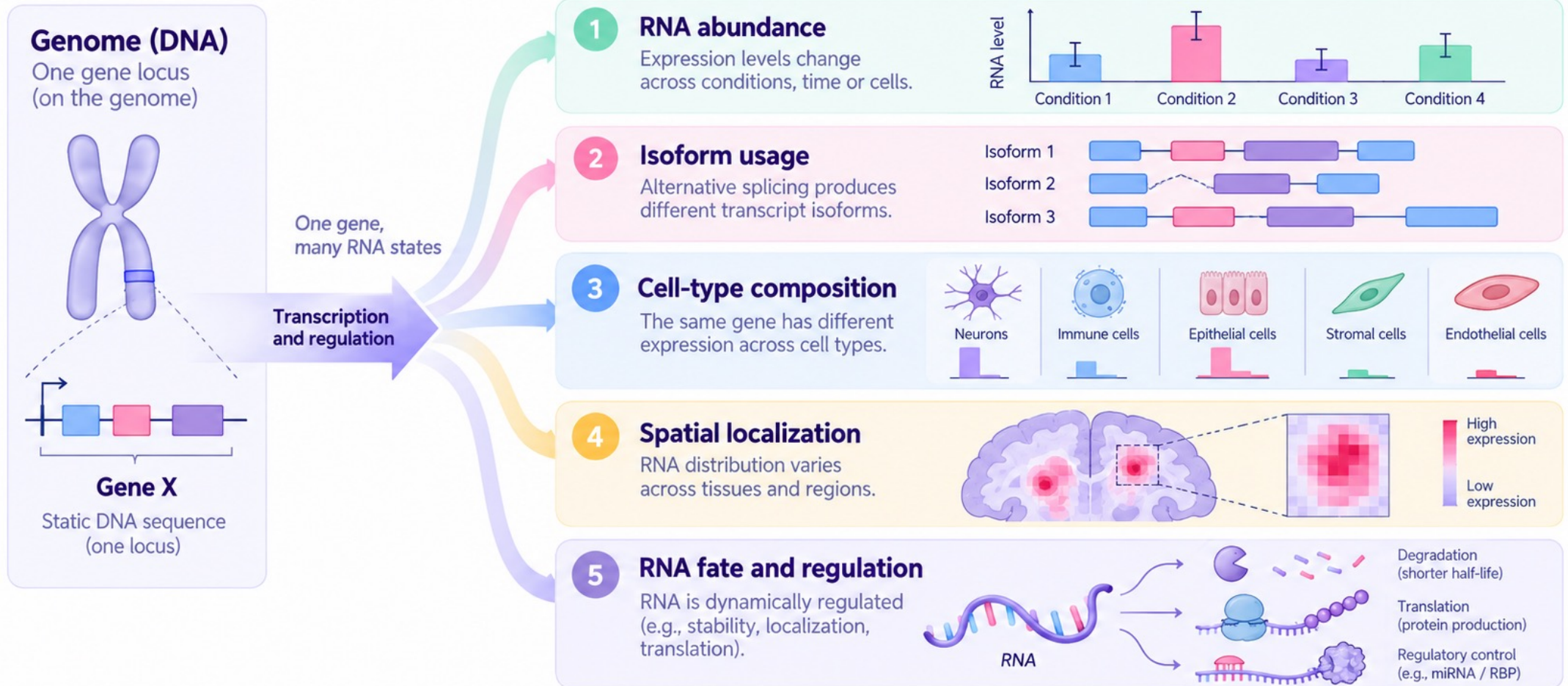


A credible transcriptomic claim links biology, assay, data, statistics, and validation.

The transcriptome is dynamic — the genome is not enough

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One genome can give rise to many RNA states across cell types, spaces, and conditions.



Transcriptomics captures **changing RNA states** that the static genome alone cannot explain.

RNA-seq does not directly measure protein or mechanism

Week 5 | Slide 5

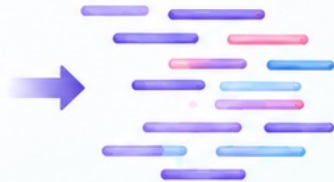
mRNA levels are a starting point, but multiple layers separate RNA from protein and phenotype.

RNA abundance

What RNA-seq measures



RNA-seq



Sequencing reads
(aligned to genes)

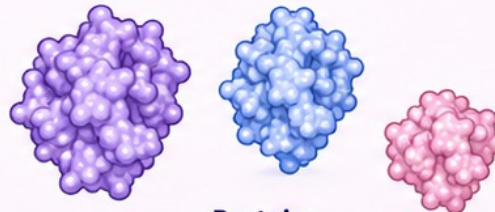


Measured directly

Quantifies mRNA abundance
(genes or transcripts).

Protein abundance

What is produced in the cell



Proteins

(abundance, isoforms, localization)

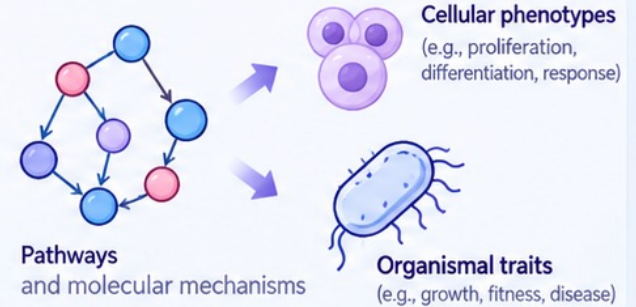


May correlate

mRNA levels can correlate with
protein levels, but the relationship
is variable and context-dependent.

Phenotype / mechanism

What the cell does



Requires orthogonal evidence

Changes in mRNA do not prove a change
in protein levels or a causal mechanism.
Additional experiments are needed.

Lower uncertainty

Higher uncertainty

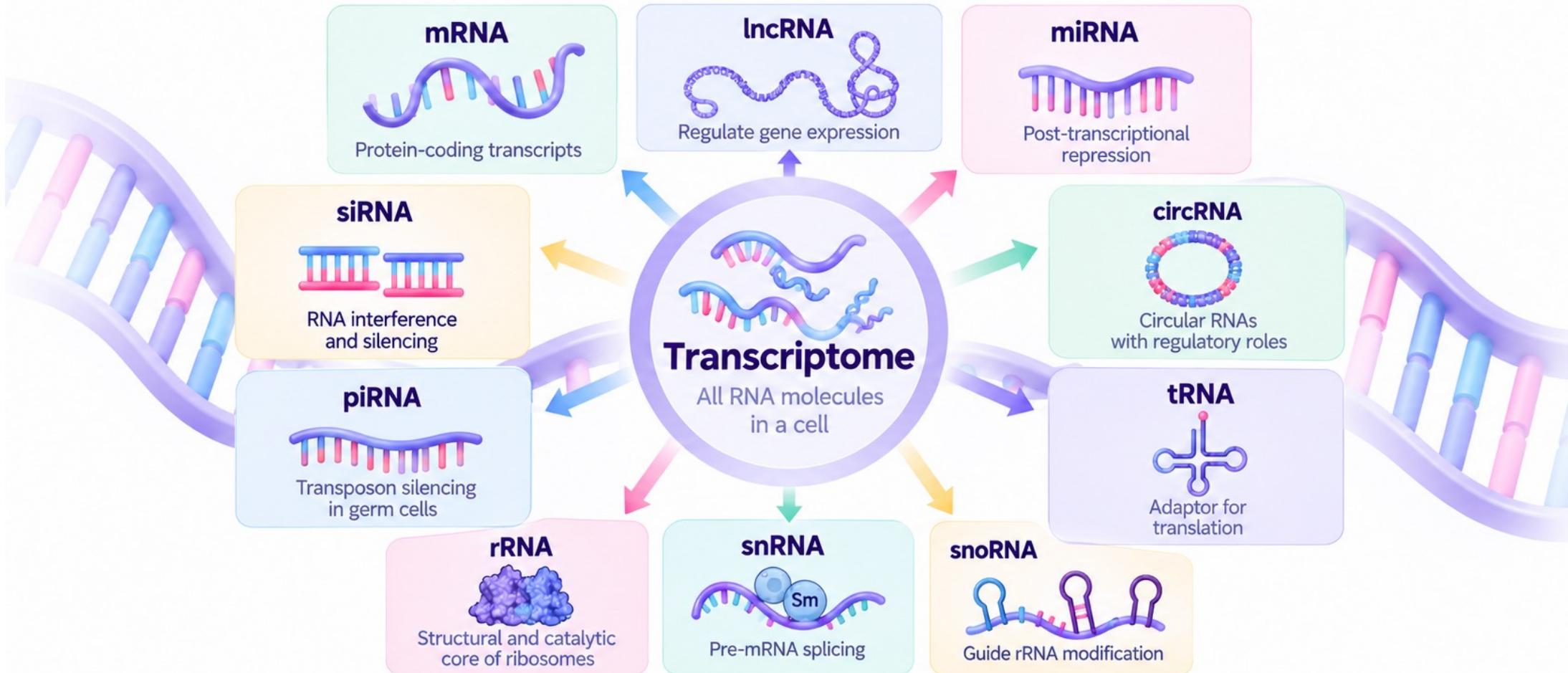


Expression changes are **evidence, not proof** of protein change or causal mechanism.

The RNA universe extends far beyond mRNA

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A diverse RNA ecosystem with distinct biogenesis, structures, and functions.

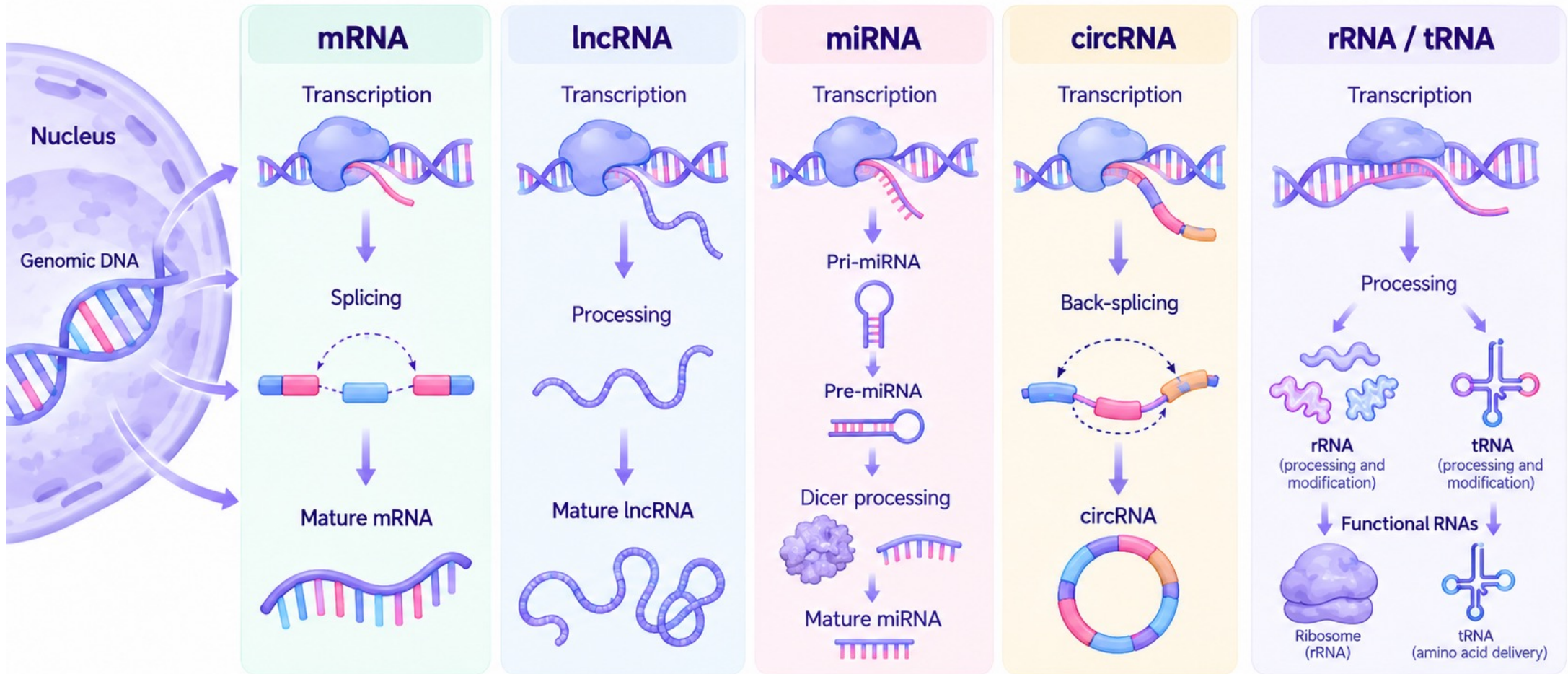


Different RNA classes carry different biological functions and require **different analytical strategies**.

How are major RNAs made?

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Different biogenesis pathways generate diverse RNA molecules from genomic DNA.

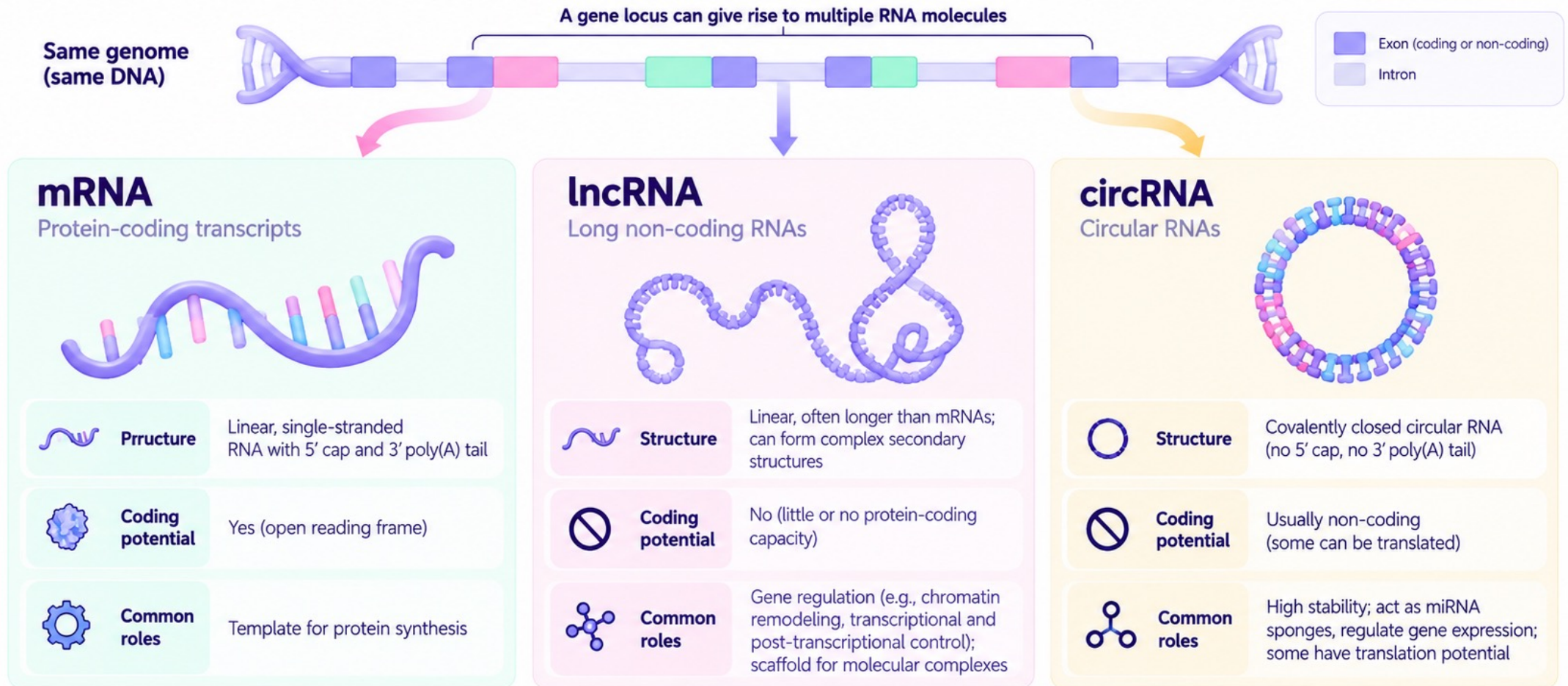


Different RNA classes arise through **distinct biogenesis pathways** that shape what sequencing can capture.

mRNA, lncRNA and circRNA: same genome, different molecular logic

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Different RNA molecules can be produced from the same genomic DNA, but they have distinct structures and functions.



Transcript identity is defined by **structure**, **coding potential**, and **regulatory role**.

Small RNAs operate through different molecular machinery

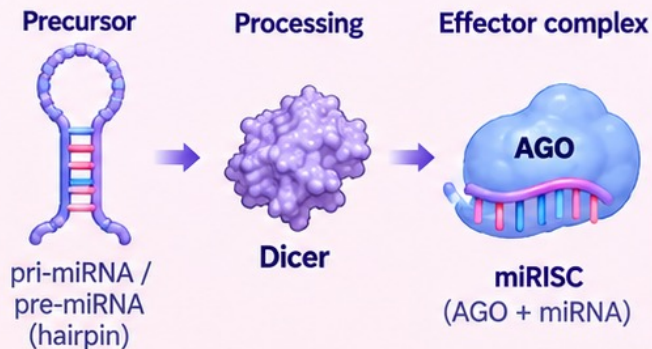
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Distinct biogenesis pathways, effector complexes, and biological roles.

miRNA

~21–23 nt

Endogenous regulation of gene expression



Main role



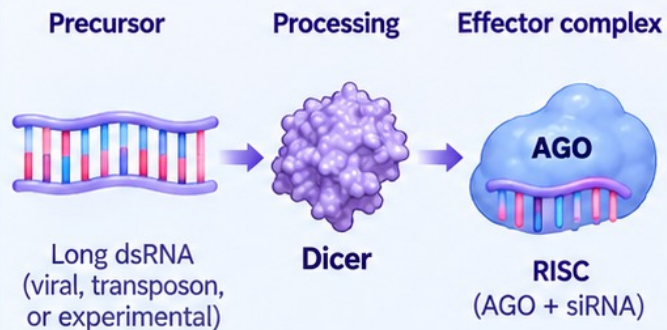
Translational repression

Inhibits protein production, often with partial complementarity.

siRNA

~21–23 nt

Defense and experimental gene silencing



Main role



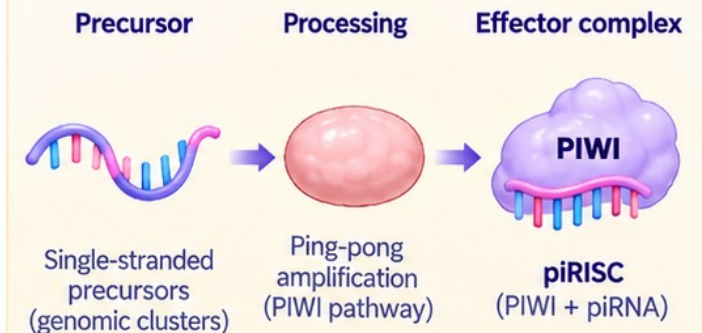
mRNA cleavage and silencing

Directs endonucleolytic cleavage of complementary transcripts.

piRNA

~24–31 nt

Transposon silencing in the germline



Main role



Transposon control

Silences transposable elements in germ cells, promoting genome stability.



Small RNAs share silencing logic, but **differ in origin, effector proteins, and biological roles.**

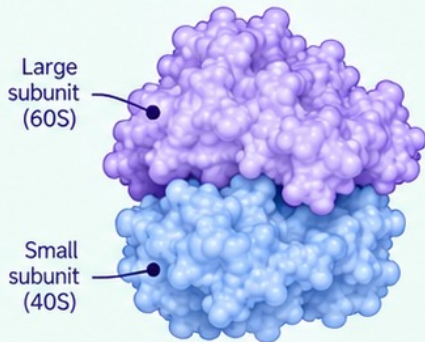
Structural RNAs are also transcriptomic information

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Beyond mRNA and regulatory RNAs, many structural RNAs are essential cellular components and are captured (to varying degrees) in transcriptomic data.

rRNA

Ribosomal RNA

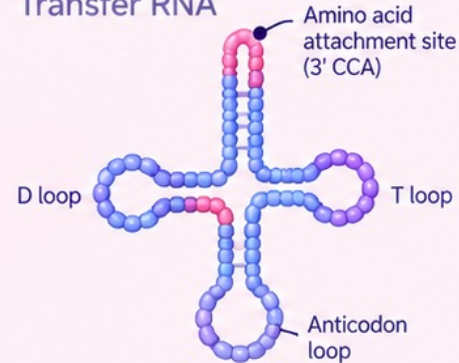


Ribosome core

Structural and catalytic core of the ribosome

tRNA

Transfer RNA

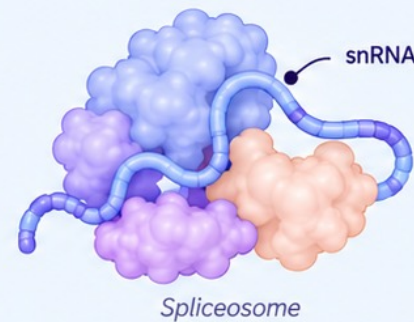


Adaptor for translation

Bridges codons and amino acids during protein synthesis

snRNA

Small Nuclear RNA

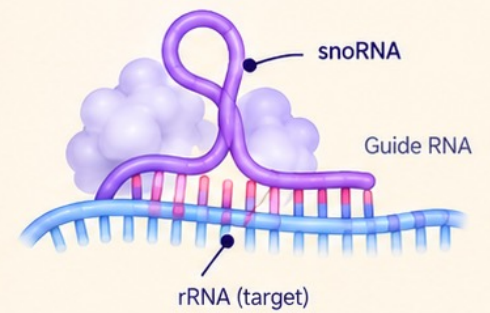


Splicing

Core components of the spliceosome for pre-mRNA splicing

snoRNA

Small Nucleolar RNA



RNA modification

Guide chemical modifications (e.g., 2'-O-methylation, pseudouridylation) of rRNAs and snRNAs



Important to note:

Standard RNA-seq may under-represent some structural RNAs (e.g., rRNAs, tRNAs, snoRNAs) due to library preparation steps and depletion strategies. **Specialized assays may be needed** for accurate detection and quantification.

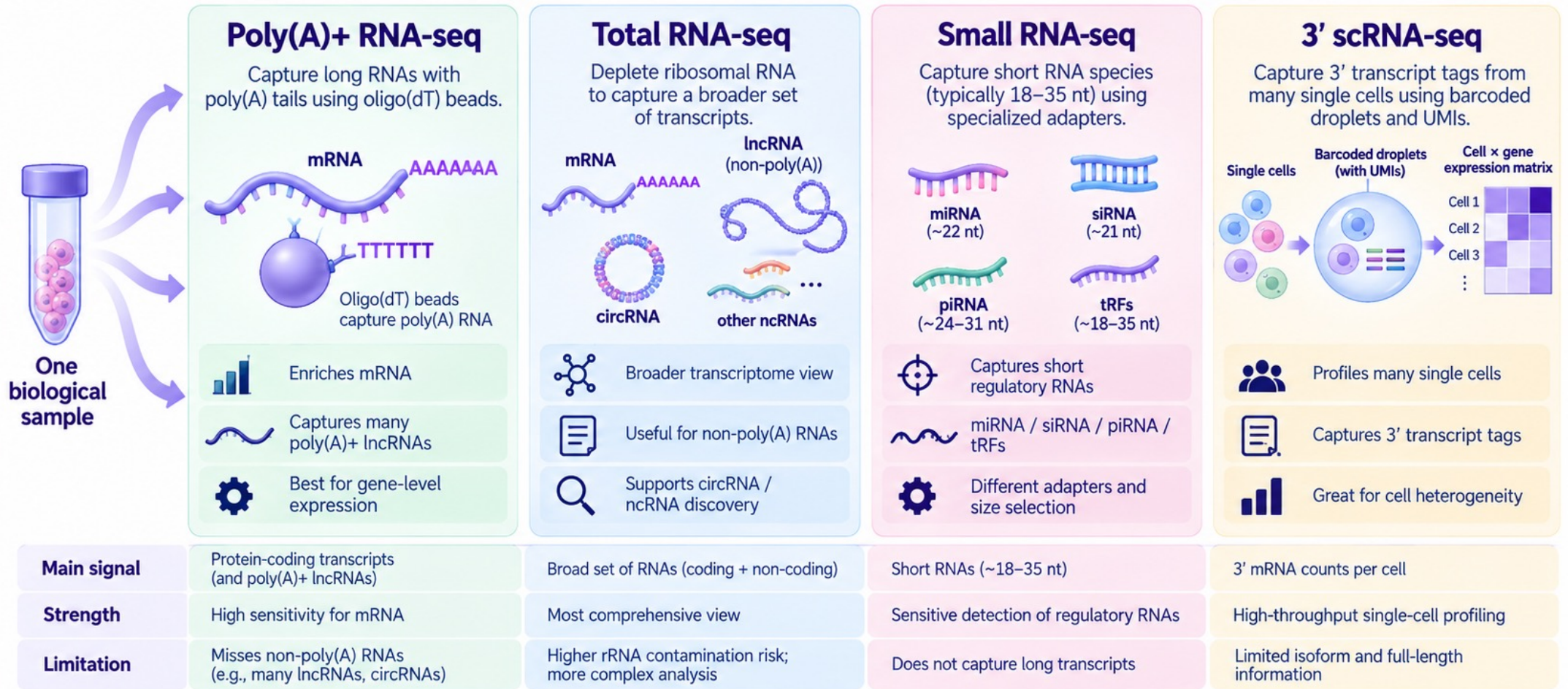


Not all transcriptomic information comes from **coding or regulatory RNAs**.

One sample, four library strategies, four different transcriptomes

Week 5 | Slide 11

Library chemistry determines which RNA molecules are captured and what biological questions can be answered.



The transcriptome you measure depends on the library you build.

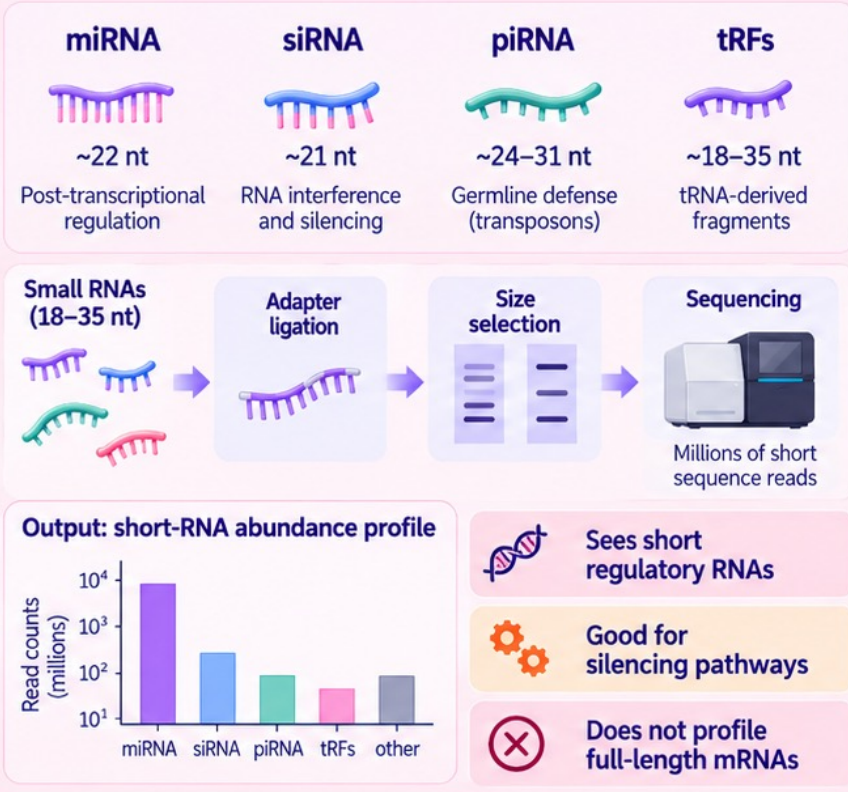
Small-RNA and single-cell libraries observe different molecular spaces

Week 5 | Slide 12

Different library designs answer different questions because they capture different molecules and different levels of biological organization.

Small RNA-seq

Profiles short regulatory RNAs from bulk cells or tissues.



What changes?

Molecule captured

Short RNAs (18–35 nt)

3' ends of mRNAs (with barcodes and UMIs)

Resolution

Population (bulk)

Single cell

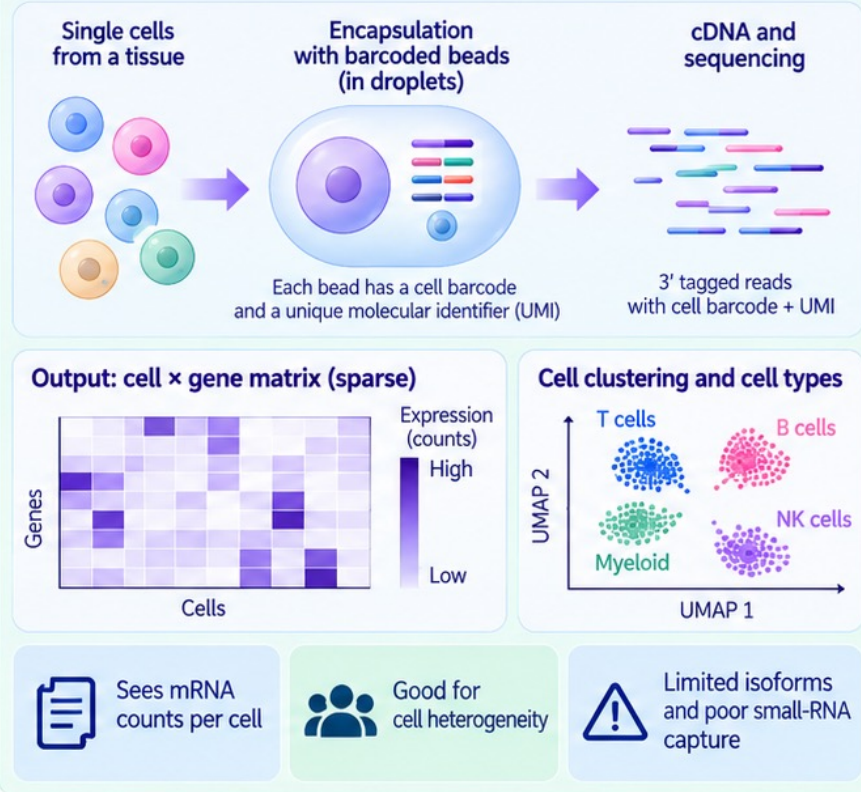
Typical question

Which small RNAs are expressed?

Which cell types are present and what genes do they express?

3' single-cell RNA-seq

Profiles 3' mRNA counts from thousands of individual cells.

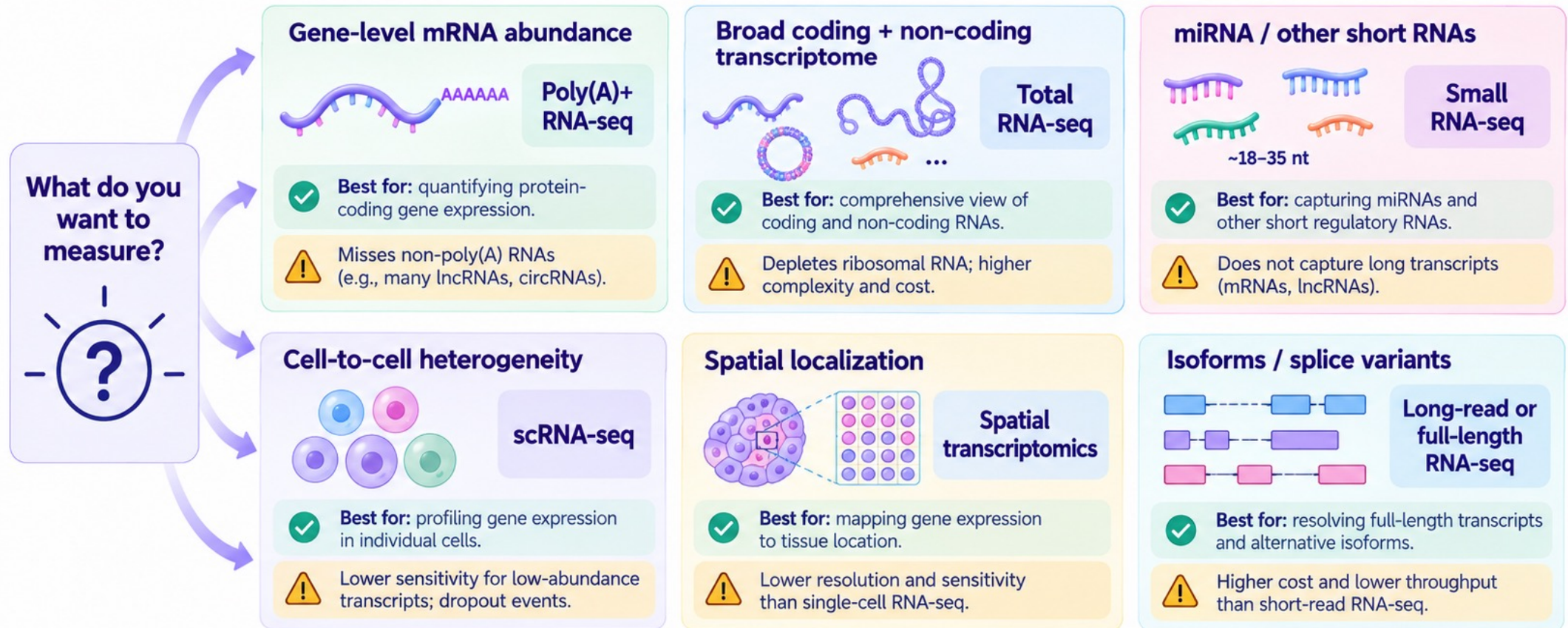


A library is not just a protocol — it defines the biological space you can observe.

Choose the assay by RNA class – not by habit

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Start from the biological question and target RNA, then match the assay to what you actually need to measure.



Bad habit to avoid

Do not default to standard bulk RNA-seq if the target RNA class or biological question requires another assay.



The right assay **follows the RNA biology**, not the researcher's habit.

Experimental design determines what statistics can estimate

Week 5 | Slide 14

Replication, balance, pairing, and batch structure decide which effects can be estimated reliably.

Balanced design

Conditions are replicated across batches.

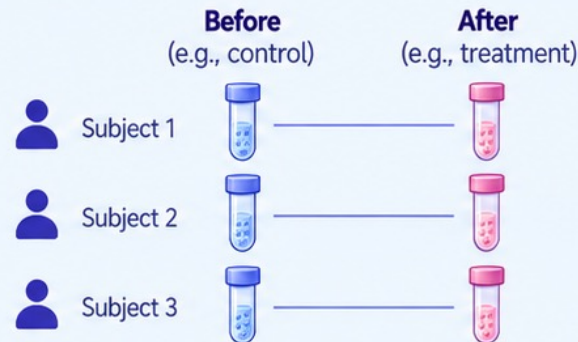
	Batch 1 (e.g., sequencing run 1)	Batch 2 (e.g., sequencing run 2)
Condition A (3 replicates)		
Condition B (3 replicates)		



- ✓ Can estimate condition effect
- ✓ Batch can be adjusted
- ✓ Best for differential expression

Blocked / paired design

Samples are paired within subjects (or matched pairs).



- ✓ Controls subject-to-subject variability
- ✓ Use paired model
- ✓ Improves sensitivity

Confounded design

Condition is perfectly aligned with batch (e.g., all controls in batch 1, all treatments in batch 2).

	Batch 1 (e.g., sequencing run 1)	Batch 2 (e.g., sequencing run 2)
Control (3 replicates)		—
Treatment (3 replicates)	—	



- ✗ Condition is confounded with batch
- ✗ Cannot separate effects
- ✗ Statistics cannot rescue poor design

What the model needs

These pieces of information must be recorded in the experimental design and metadata.



Condition

Which biological condition (e.g., control vs treatment).



Replicate

Independent biological replicates.



Batch

Technical batch (e.g., sequencing run, library prep).



Pair / subject

Subject identity or paired/matched samples (if applicable).

Missing or confounded metadata (e.g., no replicates, no batch labels, or condition aligned with batch) breaks the ability to estimate the corresponding effects.



Good models **cannot recover information** that was never created by the experiment.

What is a biological replicate?

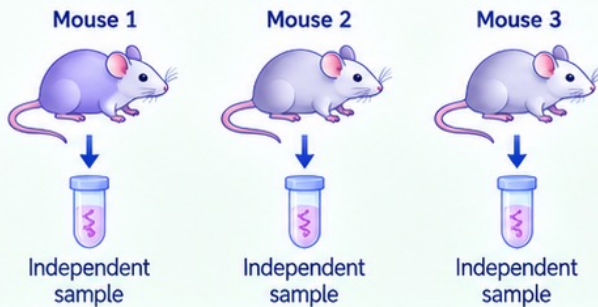
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A biological replicate is an independently obtained biological unit — not multiple measurements of the same unit.



True biological replicates

Independent biological units, each producing one sample.



Different animals

e.g., three mice from the same strain



Different patients

e.g., three independent human donors



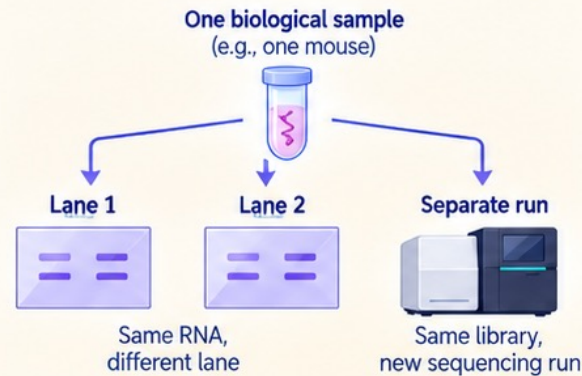
Independent cultures started separately

e.g., three cell cultures initiated from separate stocks



Technical replicates

Repeated measurements of the same biological sample (not independent units).



Same RNA, different lane

e.g., run the same sample in two lanes on a gel or two PCR reactions



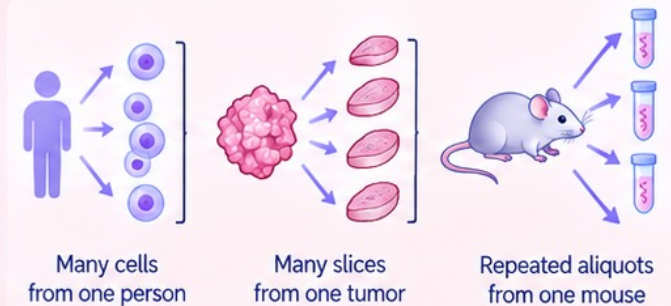
Same library, repeated measurement

e.g., sequence the same library twice or repeat quantification



Pseudoreplicates

Multiple measurements from the same biological unit, incorrectly treated as independent.



Many cells from one person

e.g., single cells treated as independent replicates



Many slices from one tumor

e.g., spatial sections from the same tumor



Repeated aliquots from one mouse

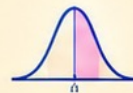
e.g., splitting one RNA sample into multiple tubes

Statistical consequence



Biological replicates estimate variance

Capture true biological variability between independent units.



Technical replicates assess measurement noise

Quantify technical variability of the same sample.



Pseudoreplicates inflate false confidence

Underestimate variability and increase the risk of false positives.

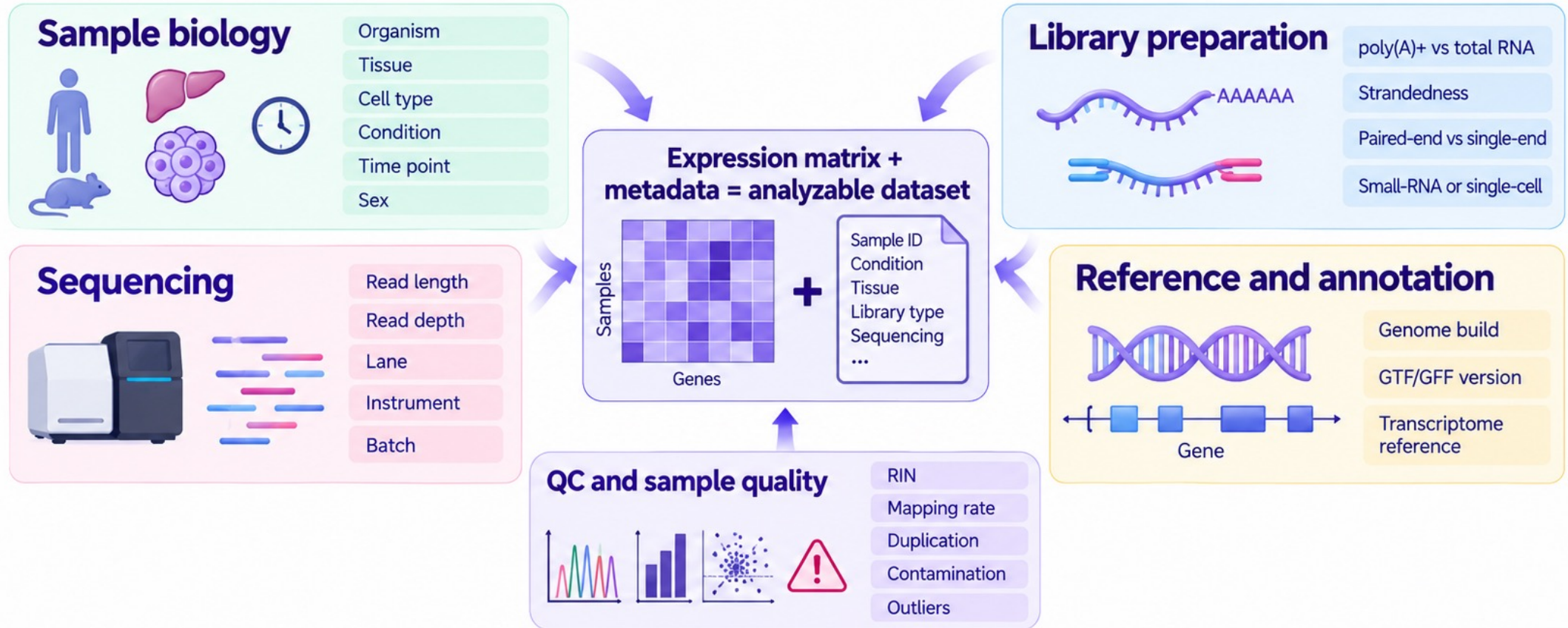


The **unit of replication** must match the **unit of inference**.

Library metadata are part of the data

Week 5 | Slide 16

Without metadata, raw sequencing files and count matrices cannot be interpreted or modeled correctly.



Missing metadata creates ambiguity in both biology and statistics.



For transcriptomics, metadata are not optional annotation — they are **part of the dataset**.

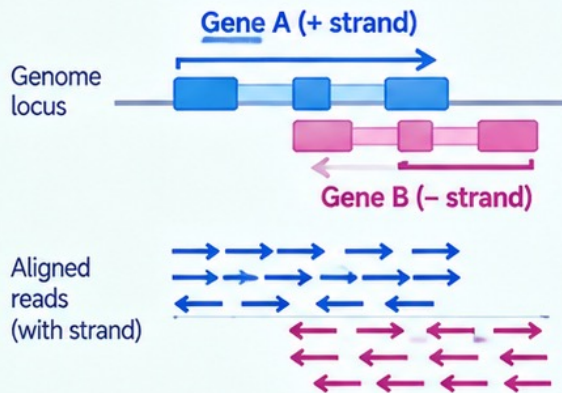
Stranded vs unstranded: the same reads can mean different genes

Week 5 | Slide 17

Read orientation matters when transcripts overlap on opposite strands or when antisense transcription is present.

Stranded library

Reads retain their orientation, so overlapping transcripts can be distinguished.



Read assignment result (stranded)

Reads assigned to Gene A: 120
Reads assigned to Gene B: 110

↔ **Orientation preserved**
Keeps track of which strand each read came from.

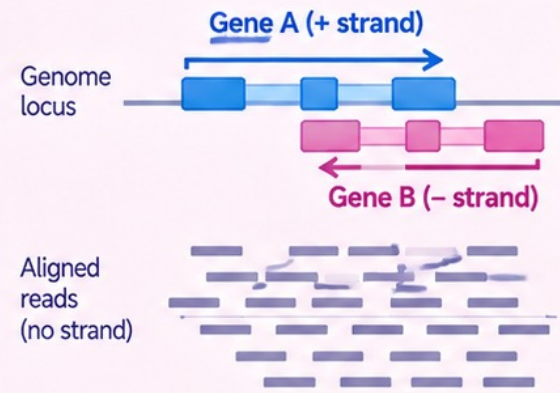
▒ **Improves assignment**
Accurately separates overlapping genes and antisense transcripts.

⚙ **Useful for antisense / overlapping loci**
Essential when transcripts overlap or are on opposite strands.

Same genomic region, different interpretability

Unstranded library

Reads do not have orientation information, making it difficult to assign reads when transcripts overlap.



Read assignment result (unstranded)

Reads assigned to Gene A: 115
Reads assigned to Gene B: 115

⊘ **Orientation lost**
Cannot determine which strand each read came from.

? **Ambiguous overlap**
Reads in overlapping regions cannot be confidently assigned to Gene A or Gene B.

▒ **Can blur gene-level interpretation**
May inflate counts, mask antisense transcription, or complicate differential expression analysis.



Practical note: Always record whether the library is forward-stranded, reverse-stranded, or unstranded before quantification.

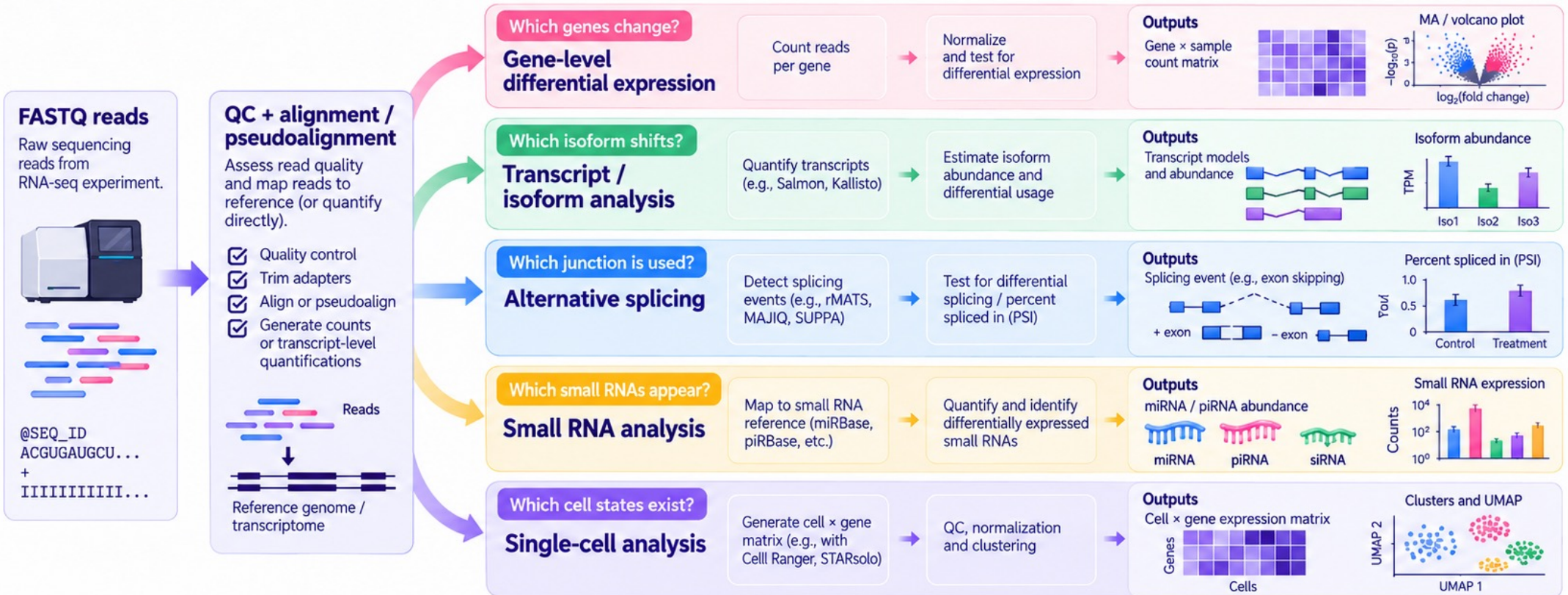


Some RNA-seq information is carried by **sequence orientation**, not only by read count.

There is no single RNA-seq pipeline

Week 5 | Slide 18

The analysis path depends on the assay, the biological question, and the level of output you want.



Why pipelines differ

Different choices are appropriate for different experiments and questions.



Library type
(e.g., poly(A)+, total RNA, small RNA, single-cell)



Reference
(genome vs. transcriptome; species and version)



Annotation
(gene models, transcript models, small RNA databases)



Statistical goal
(gene-level, isoform-level, splicing, cell states, discovery vs. test)

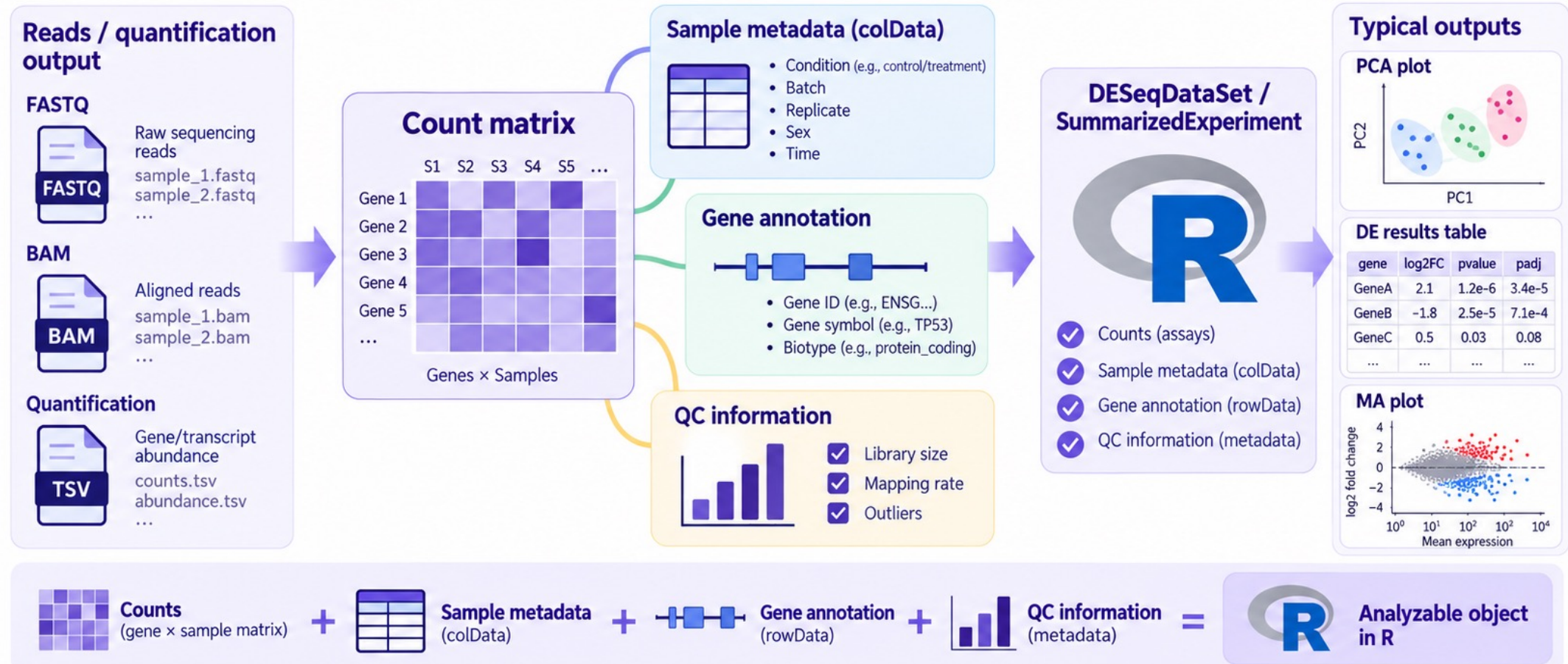


The best pipeline is the one that **matches the assay and the question.**

What object enters R?

Week 5 | Slide 19

Statistical analysis in R starts from structured data objects, not from raw sequencing images or scattered files.



R works best when transcriptomic data are organized into explicit, documented objects.

Class 1 challenge: Is this dataset analyzable?

Week 5 | Slide 20

Before touching any software, ask whether the dataset, metadata, and assay are sufficient for the biological question.

A Bulk RNA-seq, 4 vs 4, balanced batches, clear metadata

Gene count matrix
(raw counts)

Gene	S1	S2	S3	S4	S5	S6	S7	S8	...
TP53	120	98	112	101	210	195	205	198	...
BRCA1	52	61	48	55	120	130	118	125	...
GAPDH	500	520	480	510	530	515	505	525	...
...	...	...	...	...	...	...	...	...	...

Sample metadata

Sample	Condition	Batch
S1	Control	Batch 1
S2	Control	Batch 1
S3	Control	Batch 2
S4	Control	Batch 2
S5	Treated	Batch 1
S6	Treated	Batch 1
S7	Treated	Batch 2
S8	Treated	Batch 2



YES

Why?

- 1 Replicated (4 vs 4)
- 2 Estimable condition effect (with balanced batches)

B Question: miRNA changes. Data: Poly(A)+ RNA-seq only

Biological question

Differential expression of microRNAs (miRNAs)



≠

Available data / assay

Poly(A)+ RNA-seq (mRNA-focused)



MAYBE / NO

Why?

- 1 Wrong assay for miRNAs (Poly(A)+ RNA-seq does not capture most miRNAs)
- 2 Need small RNA-seq or another assay (e.g., miRNA-seq, qPCR, or targeted panels)

C Control samples in Batch 1, treated samples in Batch 2 only

Sample metadata

Sample	Condition	Batch
C1	Control	Batch 1
C2	Control	Batch 1
C3	Control	Batch 1
C4	Control	Batch 1
T1	Treated	Batch 2
T2	Treated	Batch 2
T3	Treated	Batch 2
T4	Treated	Batch 2

Complete confounding

Batch 1
(all controls)



Batch 2
(all treated)



NO

Why?

- 1 Condition confounded with batch (cannot separate treatment from batch)
- 2 Cannot estimate treatment effect reliably



Before you analyze, ask yourself:

1

What is the biological question?

2

What assay was used?

3

What information is missing?

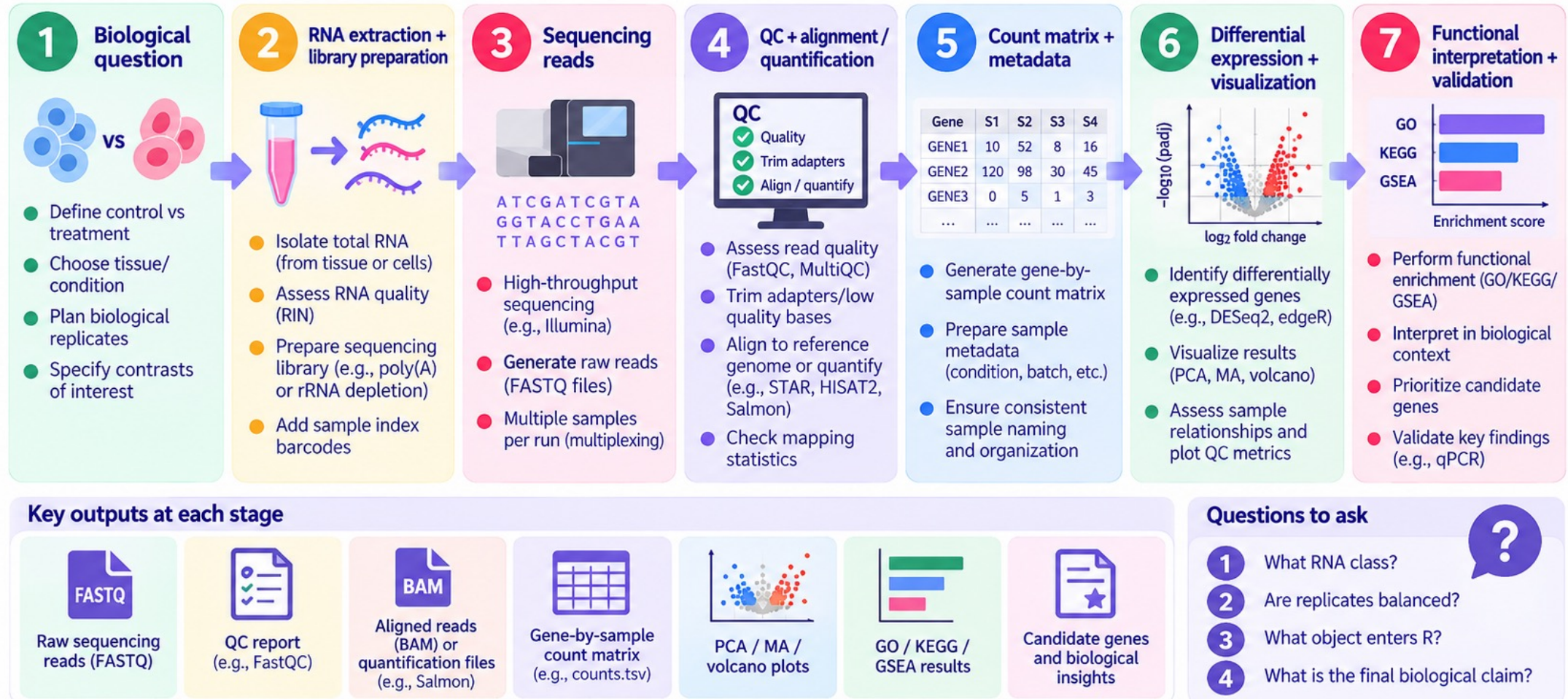


Dataset analysis **begins with judgment**, not with clicking Run.

From RNA to insight: the bulk RNA-seq analysis roadmap

Week 5 | Slide 21

A successful analysis moves step by step from biological question to interpretable results.



Good transcriptomics is a **workflow**, not a single button click.

Quality control asks whether the data deserve interpretation

Week 5 | Slide 22

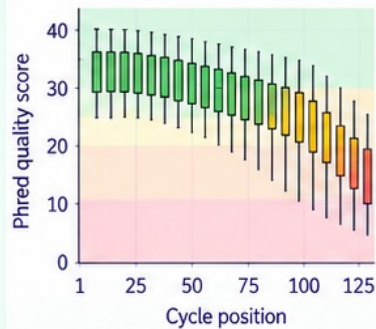
QC is not decoration — it tells you what can be trusted, what must be fixed, and what should be excluded.

1

Raw read quality

Are base calls reliable across read length?

Per-base sequence quality



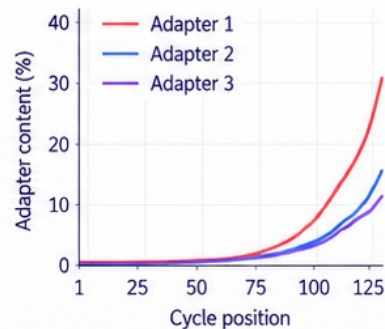
Low tail quality suggests trimming.

2

Adapter contamination

Are sequencing adapters present in the reads?

Adapter content by cycle



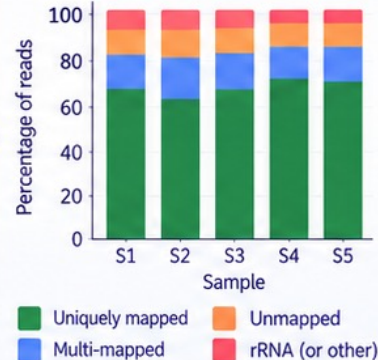
Trim adapters before alignment.

3

Alignment / mapping summary

Do reads map as expected?

Mapping summary per sample



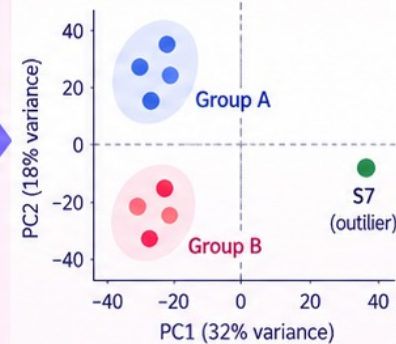
Low mapping may indicate contamination or wrong reference.

4

Sample-level consistency

Do samples cluster as expected?

PCA of samples



Check whether biology or technical artifact explains separation.

5

Outlier decision

What should we do with problematic samples?



Keep

Passes QC and matches expected behavior.



Review

Potential issue. Investigate further (metadata, rerun QC).



Remove

Clear technical failure or inconsistent with the rest of the dataset.



Common QC metrics



Per-base quality

Base calling accuracy across cycles



GC content

Overall and per-sample GC distribution



Duplication

Fraction of duplicate reads



Insert size

Distribution and expected fragment size



Mapping rate

Fraction of reads that align



Gene body coverage

Uniformity across genes (e.g., 5' → 3')



QC **changes the analysis plan** before the statistics begin.

Not all RNA-seq values mean the same thing

Choose the value type that matches the analytical task.

Week 5 | Slide 23

A Raw counts
Integer gene-by-sample counts.

Gene	S1	S2	S3	S4
TP53	104	87	120	95
BRCA1	0	5	2	1
GAPDH	5234	4987	6121	5290
ACTB	1876	2011	1899	2055
...	...	...	...	...

✓ **What it is**
Integer gene-by-sample counts from the sequencer.

✓ **Used for**
Best for DESeq2 / edgeR modeling (count-based methods).

✗ **Do NOT use for**
Direct sample comparison without normalization.

B Normalized counts
Size-factor adjusted counts.

Gene	S1	S2	S3	S4
TP53	98.7	102.3	110.1	96.5
BRCA1	0.0	4.9	2.1	1.3
GAPDH	5010	5122	5903	5201
ACTB	1799	2068	1834	2017
...	...	...	...	...

✓ **What it is**
Size-factor adjusted counts (e.g., from DESeq2, edgeR, or similar).

✓ **Used for**
Useful for reporting and some heatmaps.

✗ **Do NOT use for**
Not the main input for DESeq2 hypothesis testing if produced externally (prefer raw counts).

C TPM / CPM
Length- or depth-adjusted abundance.

Gene	S1	S2	S3	S4
TP53	12.4	10.1	15.6	11.8
BRCA1	0.0	0.3	0.1	0.1
GAPDH	245.2	238.7	260.1	251.3
ACTB	87.6	92.1	80.5	90.2
...	...	...	...	...

✓ **What it is**
Length- or depth-adjusted abundance (Transcripts Per Million or Counts Per Million).

✓ **Used for**
Useful for expression comparison within or across samples, isoform abundance.

✗ **Do NOT use for**
Not ideal for DESeq2 count models (not integer counts; compositionally constrained).

D VST / rlog
Variance-stabilized transformed values.

A PCA plot with PC1 (42%) on the x-axis and PC2 (16%) on the y-axis. Four clusters of points are shown, labeled S1, S2, S3, and S4. S1 is a blue cluster at the top left, S2 is a green cluster at the bottom left, S3 is a red cluster at the top right, and S4 is a purple cluster at the bottom right.

✓ **What it is**
Variance-stabilized transformed values (e.g., DESeq2 VST or rlog).

✓ **Used for**
Useful for PCA, clustering, heatmaps.

✗ **Do NOT use for**
Not used as raw input for count-based DE tests (e.g., DESeq2, edgeR).

Which value for which task?

DE testing
(differential expression)
A Raw counts

PCA / clustering
(sample relationships)
D VST / rlog

Expression display
(tables, heatmaps)
B Normalized counts

Transcript abundance
(isoforms, within/across samples)
C TPM / CPM



The right value object depends on the question, not on habit.

Normalization removes library-size bias, not biology

Week 5 | Slide 24

Normalization makes samples comparable, but it cannot rescue poor experimental design.

A

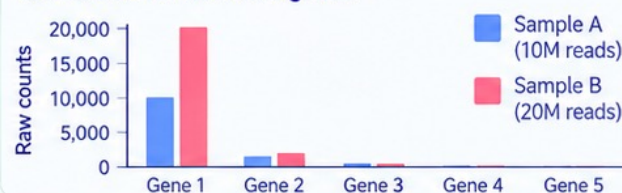
Before normalization

Different library sizes lead to misleading raw counts for the same genes.

Raw counts (subset of genes)

Gene	Sample A (total 10M)	Sample B (total 20M)
Gene 1 (highly expressed)	10,000	20,000
Gene 2 (medium)	1,000	2,000
Gene 3 (low)	100	200
Gene 4 (low)	50	100
Gene 5 (very low)	10	20

Raw counts for the same genes



Same biology, different sequencing depth makes Sample B look higher for all genes.

B

How size-factor normalization works

Scale counts by a sample-specific factor to make libraries comparable.



Example: One gene, two library sizes

Sample	Total reads (library size)	Raw count (Gene X)	Size factor	Normalized count (Gene X)
Sample A	10,000,000	100	1.0	100
Sample B	20,000,000	200	2.0	100



After normalization, the gene has similar expression (100 vs 100) and samples are comparable.

C

What changes and what does not

Normalization fixes sequencing depth differences, but it cannot fix flawed experimental design.

Changes (improves comparability)



Library-size effects decrease

Samples become comparable in scale.



Count distributions align better

Overall expression distributions are more similar.



Relative expression between samples is meaningful

True biological differences can be assessed (when design is sound).

Does not change (still a problem)



Batch effects

Technical batch/compositional effects remain.



Wrong assay choice

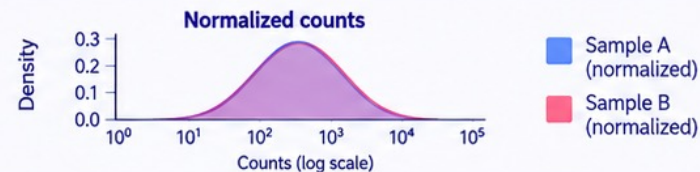
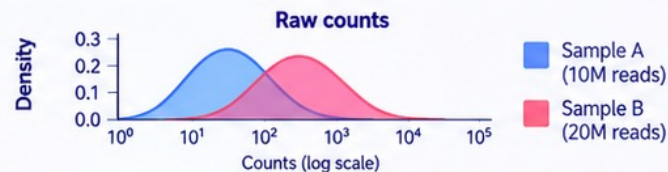
Normalization cannot recover an assay that does not measure the biology of interest.



Confounded experimental design

Normalization does not fix missing replicates, confounded groups, or other design flaws.

Expression distributions before and after normalization



Normalization does not fix missing replicates or confounding.

A well-designed experiment with replicates, proper controls, and clear group structure is essential for valid biological conclusions.

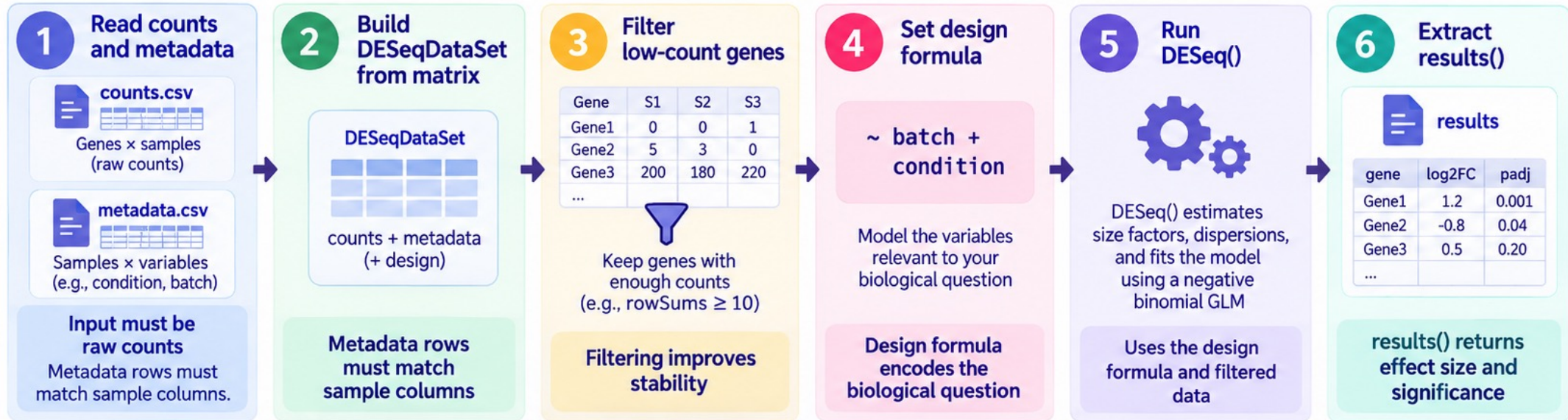


Normalize sequencing depth; do not normalize away **biological thinking**.

DESeq2 in R: from counts to a testable model

Week 5 | Slide 26

The core workflow is short, but every line depends on the design and metadata being correct.



R code

```
1 counts <- read.csv("counts.csv", row.names=1)
2 meta <- read.csv("metadata.csv")
3 dds <- DESeqDataSetFromMatrix(countData=counts,
                                colData=meta,
                                design=~ batch + condition)
4 dds <- dds[rowSums(counts(dds)) >= 10, ]
5 dds <- DESeq(dds)
6 res <- results(dds, contrast=c("condition", "Treated", "Control"))
```

} Read counts and metadata

} Build DESeqDataSetFromMatrix (with design)

} Filter low-count genes

} Run DESeq()

} Extract results()



Before you run, ask yourself:

1

Are the counts **raw** and properly formatted?

2

Do metadata rows match sample columns?

3

Does the design formula match your biological question?



R code is only correct when the **object and the **design** are correct.**

How to read a DESeq2 results table

Week 5 | Slide 27

Every column answers a different statistical question.

A

baseMean

Average normalized expression.

Mean of normalized counts across all samples. Reflects the overall expression level of the gene.

B

log2FoldChange

Effect size and direction.

Log2 fold change between conditions (e.g., treated vs control). Positive = higher in treated, negative = lower in treated.

C

lfcSE

Uncertainty of the effect size.

Standard error of the log2 fold change. Smaller values indicate more precise estimates.



Typical decision rule

- 1 Often use **padj** < 0.05.
- 2 Combine significance with **|log2FC| threshold** (e.g., $|\log_2FC| \geq 1$).
- 3 Read counts and biological context still matter (e.g., is the gene relevant to your system?).

gene	baseMean	log2FoldChange	lfcSE	stat	pvalue	padj
GeneA	452.3	2.31	0.45	5.13	2.8e-07	1.1e-05
GeneB	1200.6	-0.24	0.31	-0.77	0.442	0.78
GeneC	8.7	-3.12	1.05	-2.97	0.0029	0.041
GeneD	56.4	0.68	0.58	1.17	0.242	0.63
GeneE	2205.1	-1.05	0.28	-3.74	1.8e-04	0.0062

D

stat

Wald test statistic.

Test statistic for $H_0: \log_2FC = 0$. Larger absolute values provide stronger evidence against the null hypothesis.

E

pvalue

Unadjusted evidence.

Nominal p-value from the Wald test. Probability of observing a statistic this extreme if there is no differential expression.

F

padj

Multiple-testing adjusted significance.

P-value adjusted for multiple testing (Benjamini-Hochberg FDR). Controls the expected false discovery rate.



How to interpret



Up-regulated (higher in treated)
 $\log_2\text{FoldChange} > 0$ and significant (e.g., $\text{padj} < 0.05$)



Down-regulated (lower in treated)
 $\log_2\text{FoldChange} < 0$ and significant (e.g., $\text{padj} < 0.05$)



Not significant
 $\text{padj} \geq 0.05$ (regardless of direction)



Before you analyze, ask yourself:

1

What change do I consider biologically meaningful?

2

How much uncertainty is acceptable?

3

How does this fit with my biological context?



Differential expression is not just a p-value — it is **effect size, uncertainty, and context** together.

Exploration comes before testing

Week 5 | Slide 25

PCA and sample-distance views reveal whether samples separate by biology, by batch, or by outliers.

How to read the plot

- 1 Tight within-group clusters suggest consistency
- 2 Separation along PC1 or PC2 may reflect biology or batch
- 3 Single isolated samples deserve QC review

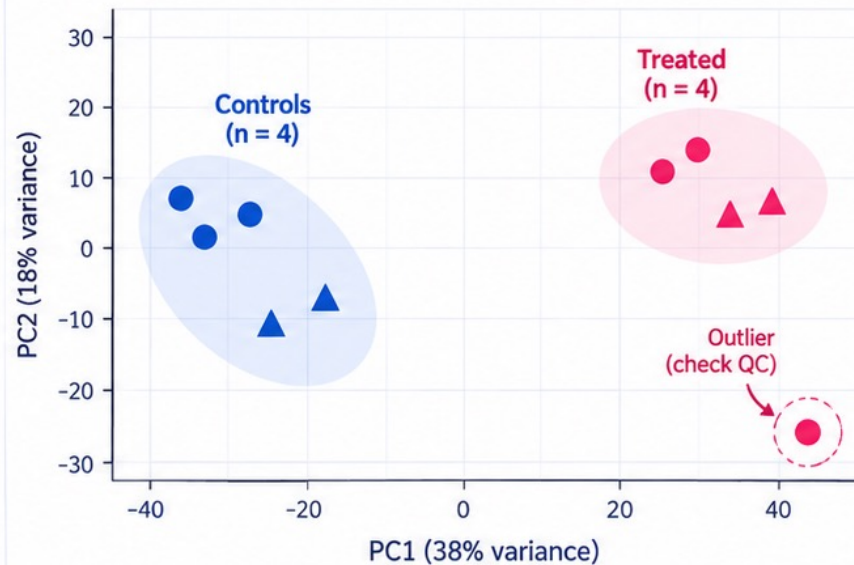
Color = condition (biology)

● Control ● Treated

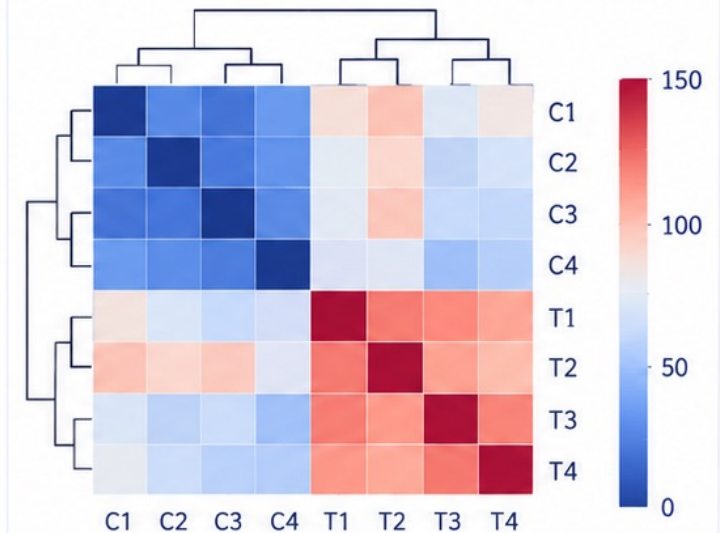
Shape = batch

● Batch 1 ▲ Batch 2

PCA of samples (RNA-seq)

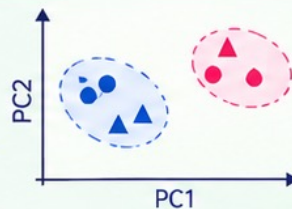


Sample-to-sample distances (Euclidean)



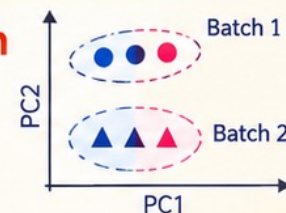
✓ Good pattern

Controls cluster together, treatments cluster together.



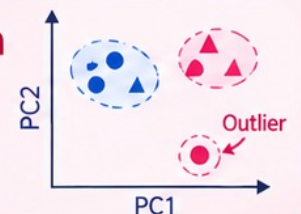
⚠ Warning pattern

Samples cluster by batch (instead of condition).



✗ Danger pattern

One or more samples are extreme outliers.

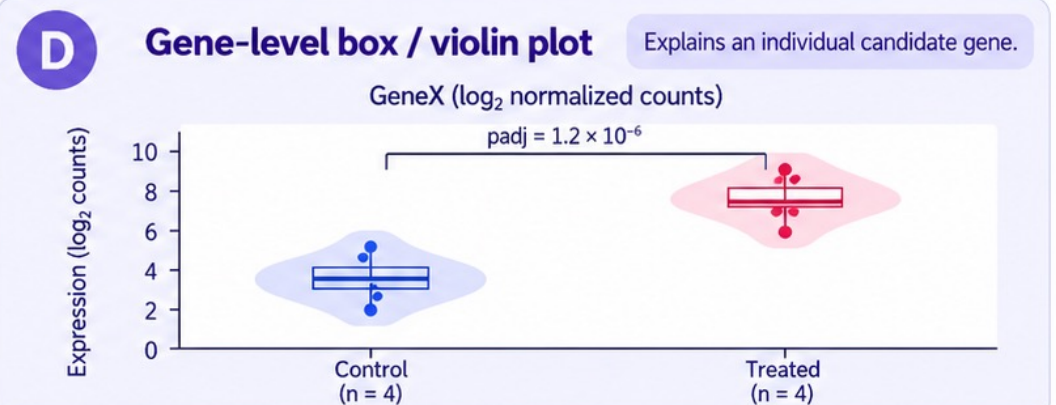
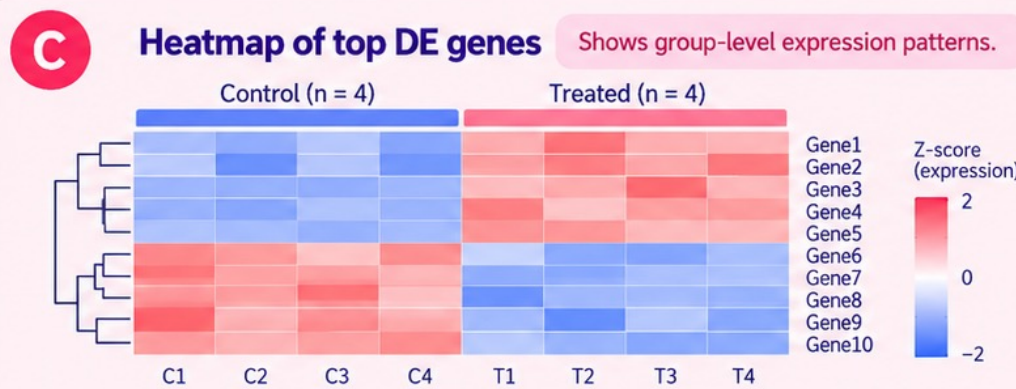
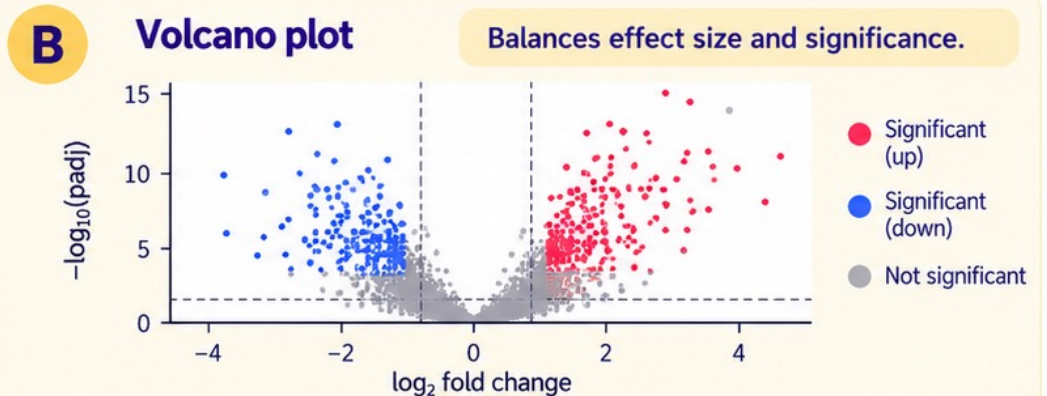
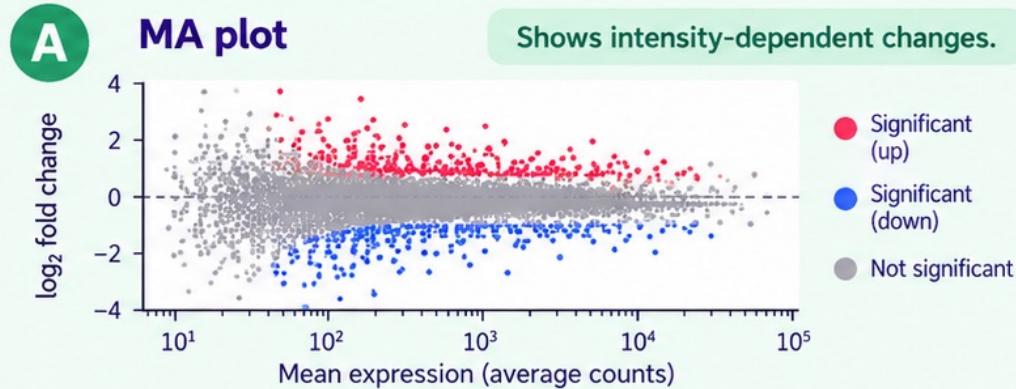


Exploratory plots tell you **what model** and **what caution** the dataset needs.

Each plot answers a different question

Use multiple plots together to interpret differential expression results.

Week 5 | Slide 28



Interpretation logic:

1

PCA first

Check sample structure and batch effects.

2

MA/volcano to screen

Find differentially expressed genes.

3

Heatmap for patterns

See group-level expression trends.

4

Gene plot for a candidate

Inspect expression and variability.



No single plot tells the whole story; the interpretation comes from **combining views**.

From DE genes to biology: ORA and GSEA answer different questions

Week 5 | Slide 29

Pathway analysis depends on whether you use a thresholded gene list or a ranked gene list.

A ORA (over-representation analysis)

Are certain pathways enriched among your significant genes?

Thresholded DE gene list
e.g., FDR < 0.05

Gene
IL6
TNF
CXCL8
CD68
NLRP3
PTGS2
...

Pathway databases

GO
(Gene Ontology)

KEGG
(Pathway maps)

Enrichment results

Pathway term	Overlap (k / K)	p-value	FDR
Inflammatory response	18 / 200	1.2×10^{-6}	2.3×10^{-4}
Mitochondrial metabolism	12 / 180	4.6×10^{-5}	6.1×10^{-3}
Cell cycle	15 / 250	1.8×10^{-4}	1.2×10^{-2}

- ✓ Input: significant gene list (e.g., FDR < 0.05)
- ✓ Simple and intuitive
- ✓ Sensitive to threshold choice

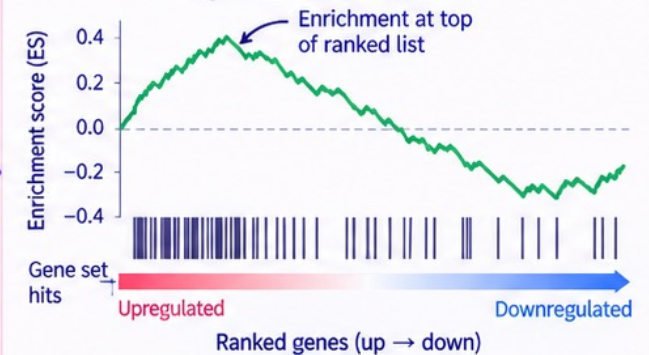
B GSEA (gene set enrichment analysis)

Are genes in a pathway coordinately up- or down-regulated?

Ranked gene list
e.g., by log2 fold change or signal-to-noise

Gene	Rank / Score
IL6	2.8
TNF	2.5
CXCL8	2.3
...	...
GAPDH	0.1
...	...
ACTB	-2.1
MT-CO1	-2.4
...	...

Running enrichment score



- ✓ Input: ranked genes (e.g., all genes with a ranking metric)
- ✓ Uses subtle coordinated changes (not just large fold changes)
- ✓ Does not require a hard significance cutoff

Category	ORA	GSEA	Key takeaway
Input	Significant gene list (thresholded)	Ranked genes (all genes)	Choose based on the structure of your evidence.
Best use case	Few strong DE genes	Distributed, modest changes across a pathway	ORA for strong hits; GSEA for subtle, coordinated shifts.
Strength	Simple and intuitive	More sensitive to coordinated changes	GSEA can detect biology missed by thresholds.
Limitation	Sensitive to threshold choice	More complex to interpret	No single method is always better.

Which method should I use?

Few strong DE genes?



Use ORA

Distributed pathway signal?



Use GSEA



Pathway analysis should match the **structure of the evidence**, not just the software menu.

AI can accelerate transcriptomics — but verification remains human

Week 5 | Slide 30

Use AI to draft code, explain outputs, and suggest next steps, but always check the biology, metadata, and statistics.



Good uses of AI

Speeds up routine tasks and helps you learn.

1



Generate R templates

Get starter code for common analyses.

2



Explain DESeq2 output

Ask for plain-language explanations of results and statistics.

3



Summarize QC metrics

Get help interpreting mapping rates, library sizes, PCA, etc.

4



Suggest visualizations

Ideas for plots (PCA, volcano, heatmap, etc.) and how to make them.

5



Draft result interpretation

Turn outputs into a clear first draft you can edit.

An agent-assisted transcriptomics workflow

Move from question to insight, with AI as a collaborator.

1

Define question



State the biological question and what data you have.

2

Ask AI for R code / workflow

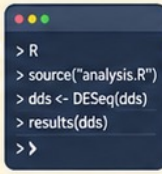


Write an R workflow for differential expression with DESeq2 ...

Use AI to propose code, explain choices, and suggest next steps.

3

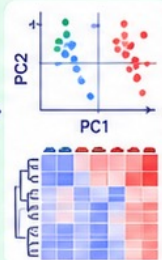
Run code on real data



Execute the code on your actual dataset (not toy examples).

4

Inspect QC and outputs



Check QC metrics, plots, and results for expected patterns.

5

Verify claims against plots, tables, and metadata

- ✓ Plots look correct
- ✓ Tables match
- ✓ Metadata consistent
- ✓ Statistics make sense
- ✓ Biology is plausible

Confirm that outputs support the claims and match the metadata.

6

Refine interpretation or redesign analysis



If something looks off, iterate: revise the question, code, or design and try again.



AI can hallucinate sample identities, code arguments, thresholds, and biological claims.

Always verify code, outputs, and interpretations against your data and trusted sources.



Checks you must still do

You are responsible for the science, even when using AI.

1



Does the assay match the RNA class?

e.g., mRNA vs. total RNA vs. small RNA.

2



Do sample labels match metadata?

Check for swapped labels, duplicates, or missing samples.

3



Is the design formula correct?

Does it match your experimental design and covariates?

4



Do plots support the claim?

Check for batch effects, outliers, and realistic effect sizes.

5



Were sources and annotations verified?

Confirm gene IDs, genome build, and external references (e.g., pathways, cell types).



Before you analyze, ask yourself:

1

What is the biological question?

2

Do the data and metadata match it?

3

What would make you trust the result?



AI is a powerful assistant, not a substitute for **experimental judgment.**

PCA: What pattern do you see?

Week 5 | Slide 31

Use PCA to inspect global sample relationships before interpreting differential expression.

Questions to interpret:



What explains PC1?

Is it treatment, cell line, batch, or something else?



Are replicates clustering?

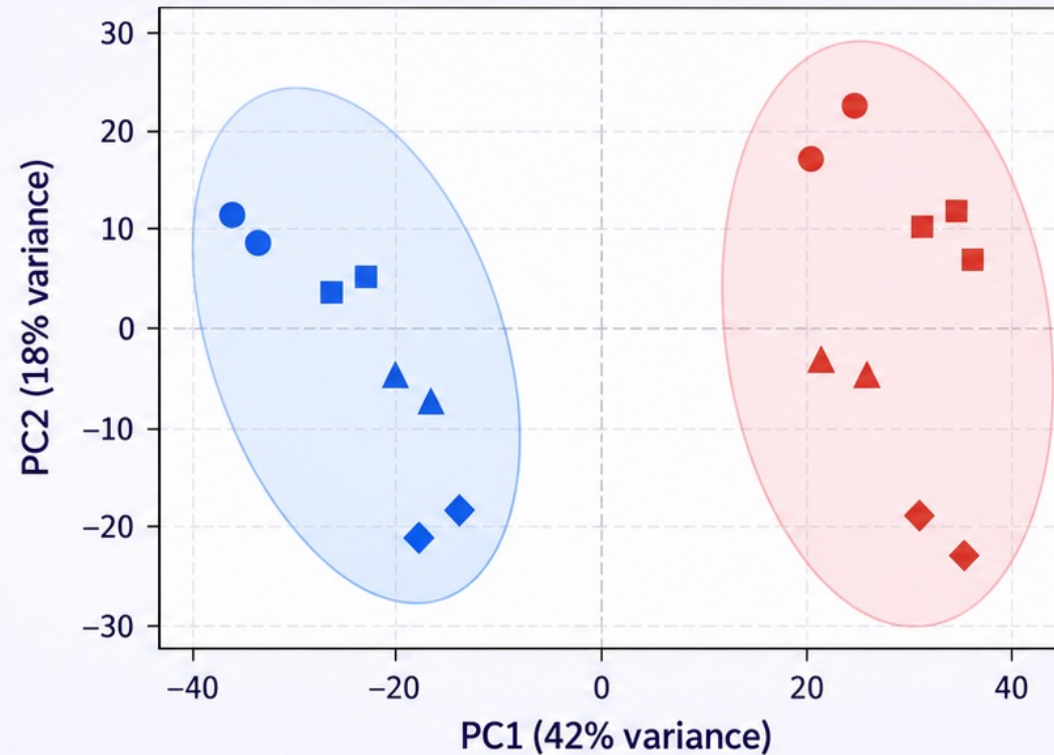
Do samples from the same condition and cell line group together?



Do you see outliers?

Are there samples that look different from the rest?

PCA of airway RNA-seq (VST)



Condition (color)

- Control
- Dex

Cell line (shape)

- A549
- BEAS-2B
- CuFi-1
- H441

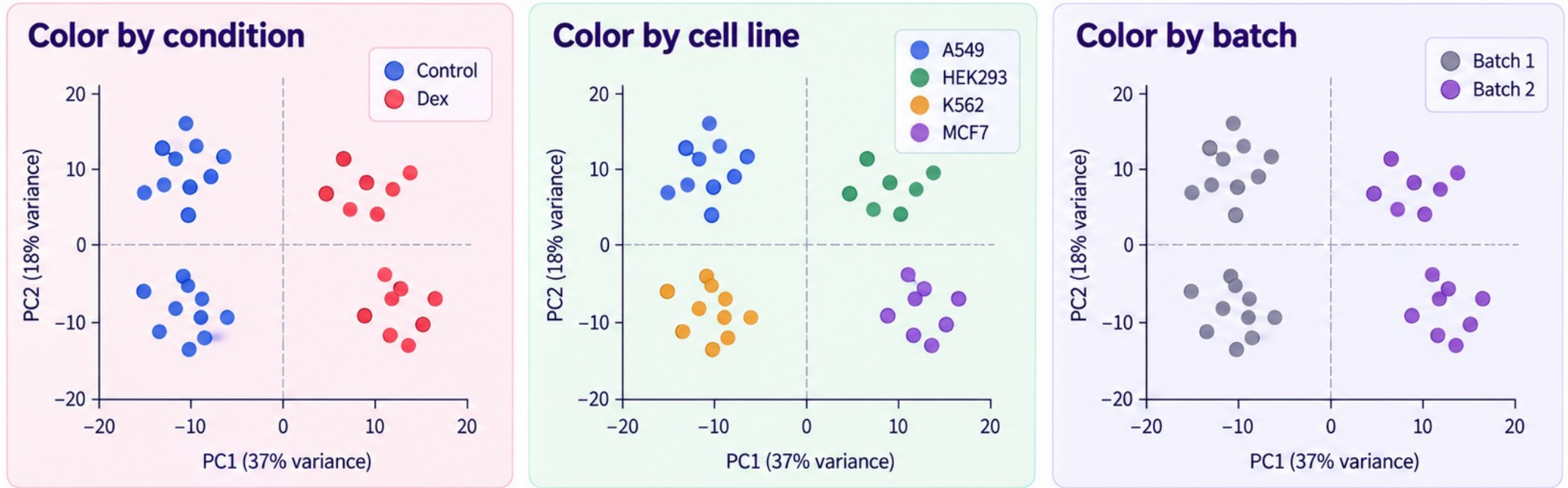


PCA shows major sources of variation, but **pattern ≠ biological cause.**

PCA interpretation: the same data, different perspectives

Week 5 | Slide 32

The same PCA can tell different stories depending on which metadata variable you highlight.



Insight

Always inspect multiple metadata variables before deciding what drives separation.

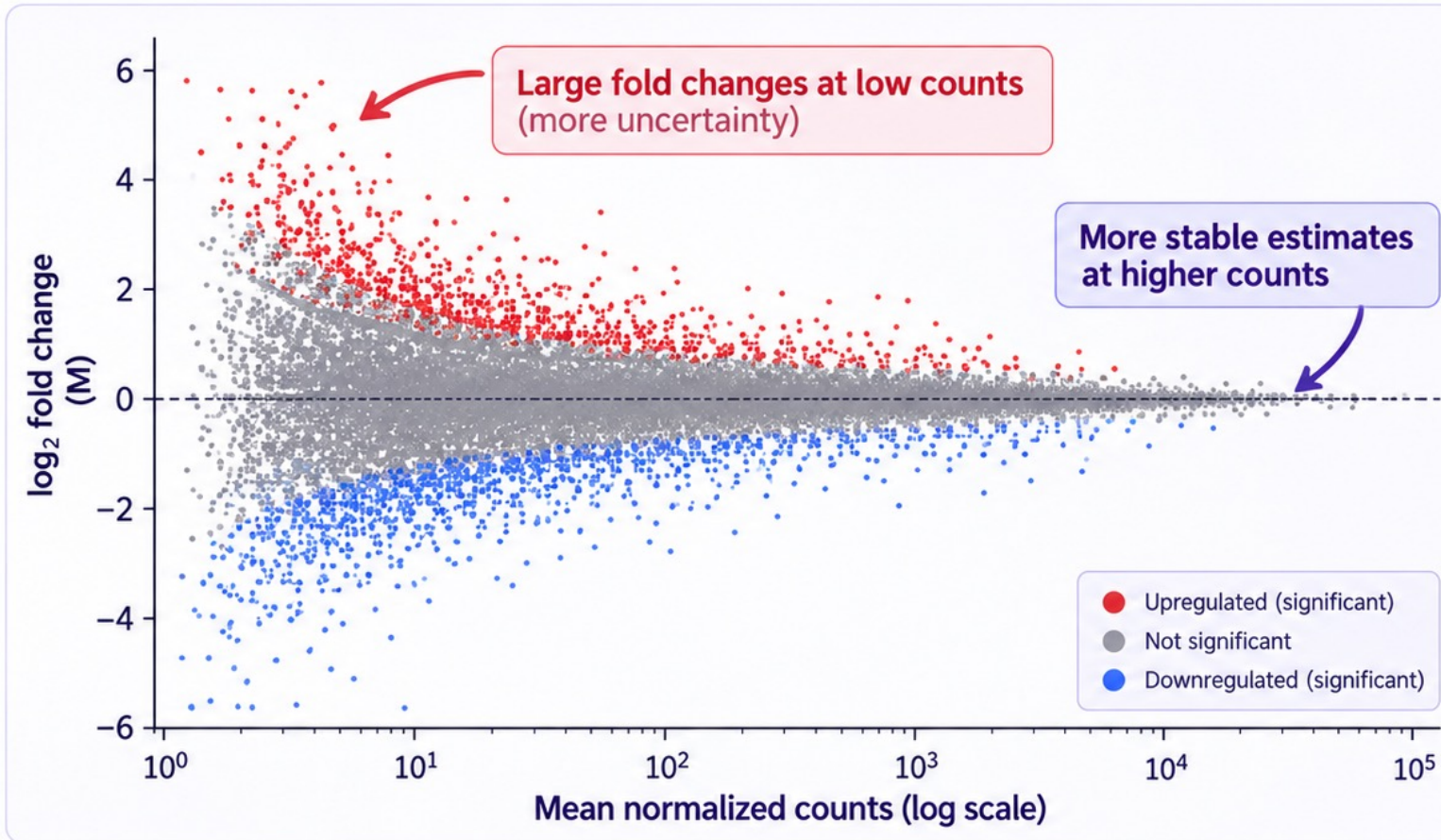


PCA reveals structure, but correct interpretation requires **biological and technical context**.

MA plot: where are the unstable effects?

Week 5 | Slide 33

Large fold changes at low counts are often less stable and require caution.



Key points



Each point is one gene.



Low counts have higher uncertainty.



Shrinkage helps reduce unstable effects.

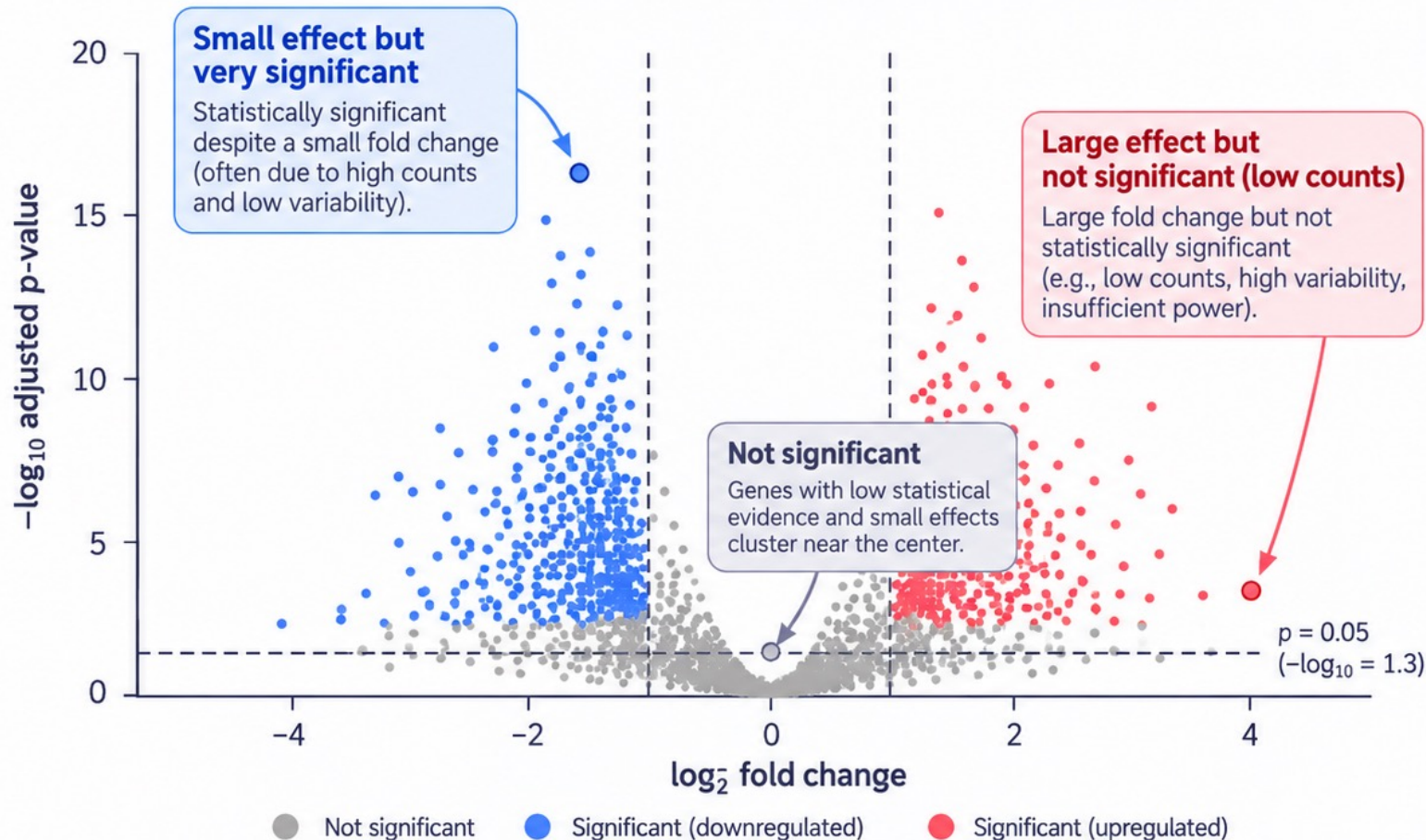


Consider **effect size** together with **mean expression** and **uncertainty**.

Volcano plot: significance is not mechanism

Week 5 | Slide 34

A gene can be statistically significant, biologically important, both, or neither.



Key takeaway



Y-axis shows statistical evidence.



X-axis shows effect size.



Interpret using **biology**, not significance alone.

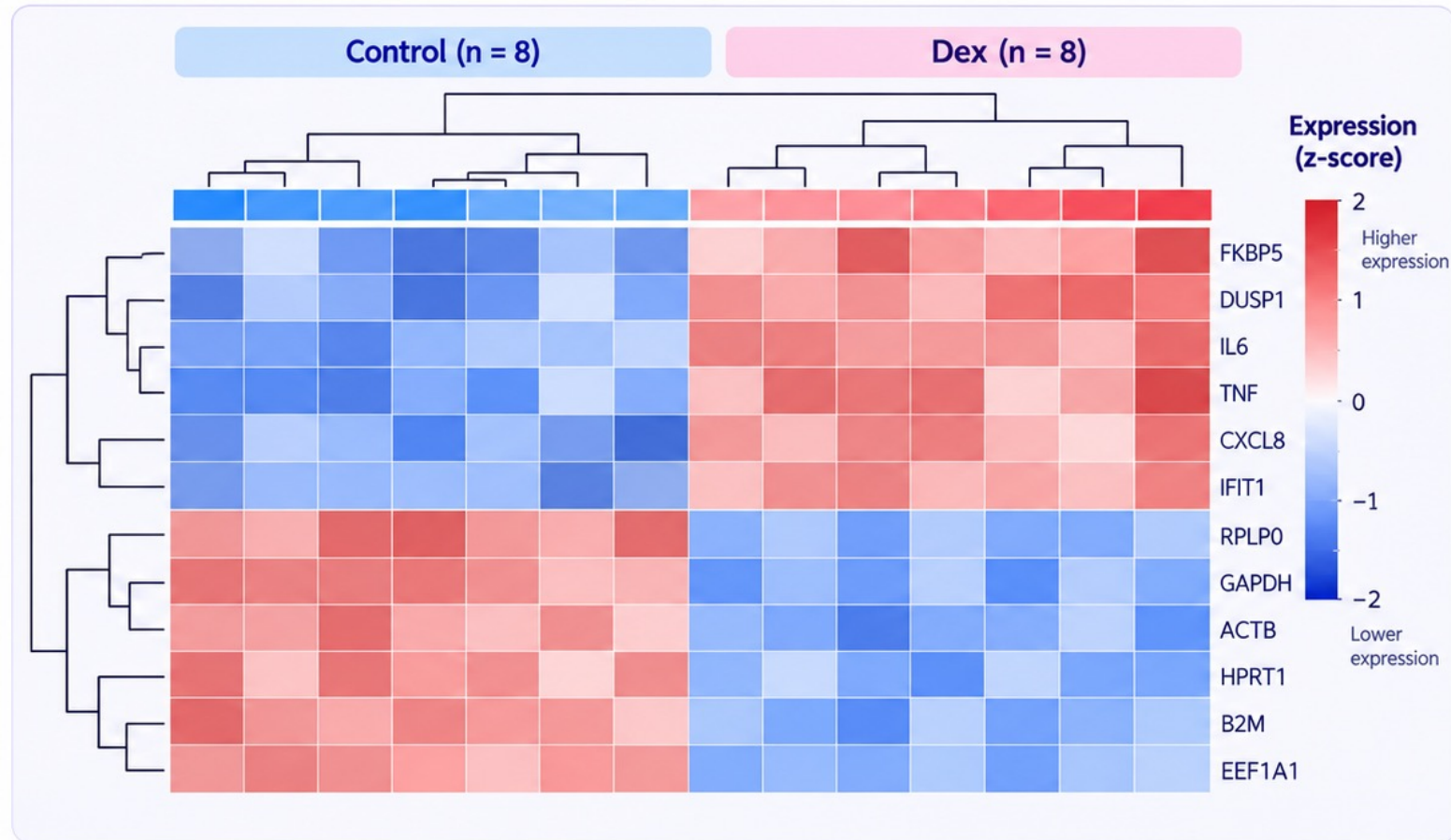


Statistical significance **does not** automatically imply biological importance.

Heatmap: why do groups appear separated?

Heatmaps show patterns, but the pattern depends on what genes and values you choose.

Week 5 | Slide 35



What affects the heatmap?

FKBP5
DUSP1
IL6
...

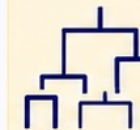
Which genes are selected

Different gene sets (e.g., DE genes, pathway genes, or all genes) can show different patterns.



What values are shown (counts, VST, z-score)

Raw counts, normalized values (e.g., VST), or row z-scores can change the appearance of the heatmap.



What distance / clustering method is used

Choices like Euclidean vs. correlation distance, and hierarchical clustering vs. k-means, can affect group separation.



Apparent separation reflects the **selected features and scaling**, not proof of mechanism.

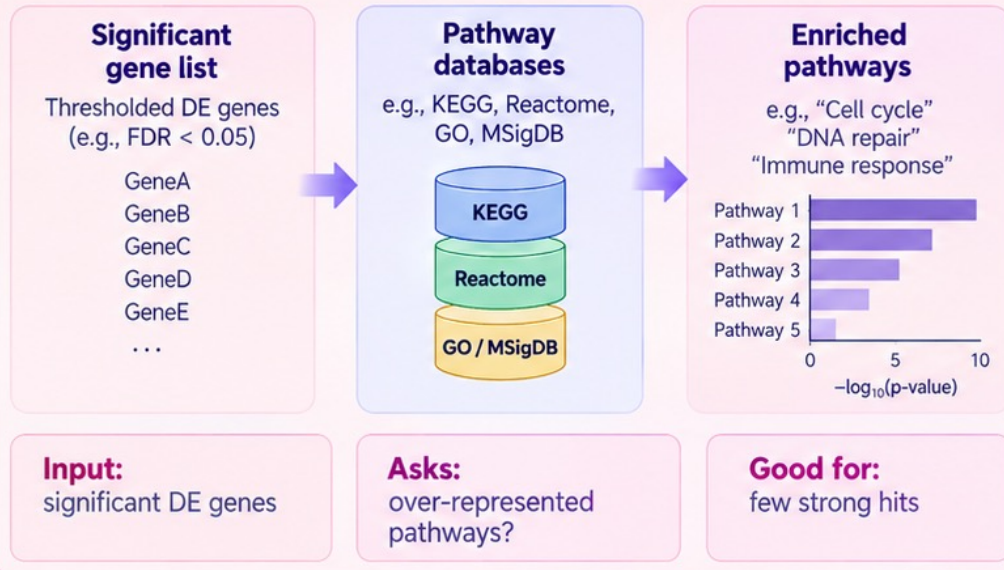
ORA vs GSEA: two different biological questions

Week 5 | Slide 36

Choose the pathway method that matches the structure of your evidence.

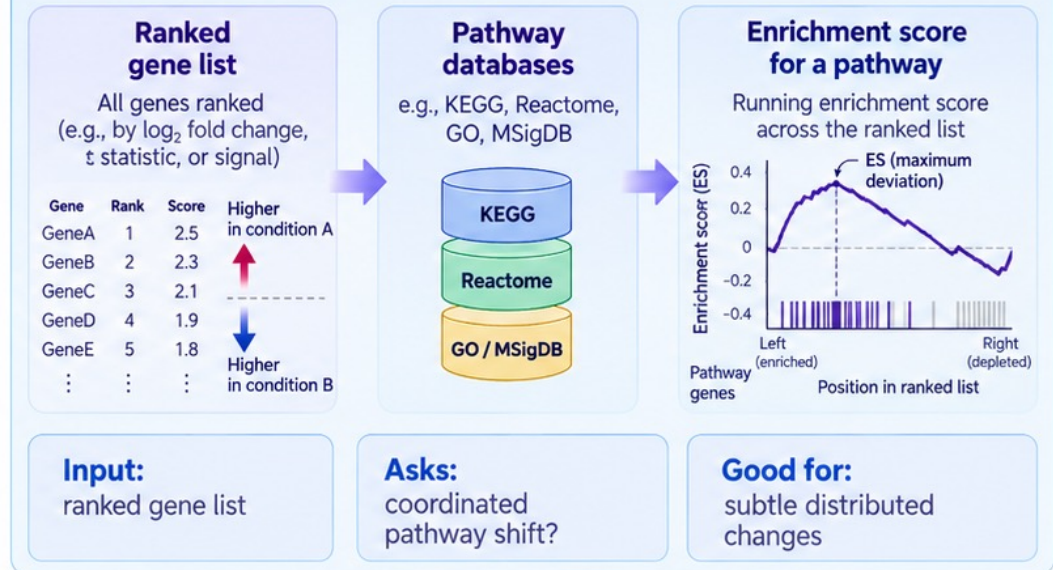
ORA (over-representation analysis)

Tests whether your list of significant genes contains more members of a pathway than expected by chance.



GSEA (gene set enrichment analysis)

Tests whether genes in a pathway are coordinately enriched at the top or bottom of a ranked gene list.



Aspect	ORA (over-representation analysis)	GSEA (gene set enrichment analysis)
Input	Significant DE genes (thresholded list)	Ranked gene list (all genes)
Strength	Simple, intuitive, fast	More sensitive, uses all genes, captures subtle coordinated changes
Typical use case	Clear, high-confidence hits (e.g., strong DE signal)	Broader, subtle, or modest changes (e.g., early responses, complex traits)



ORA and GSEA are complementary, not interchangeable.

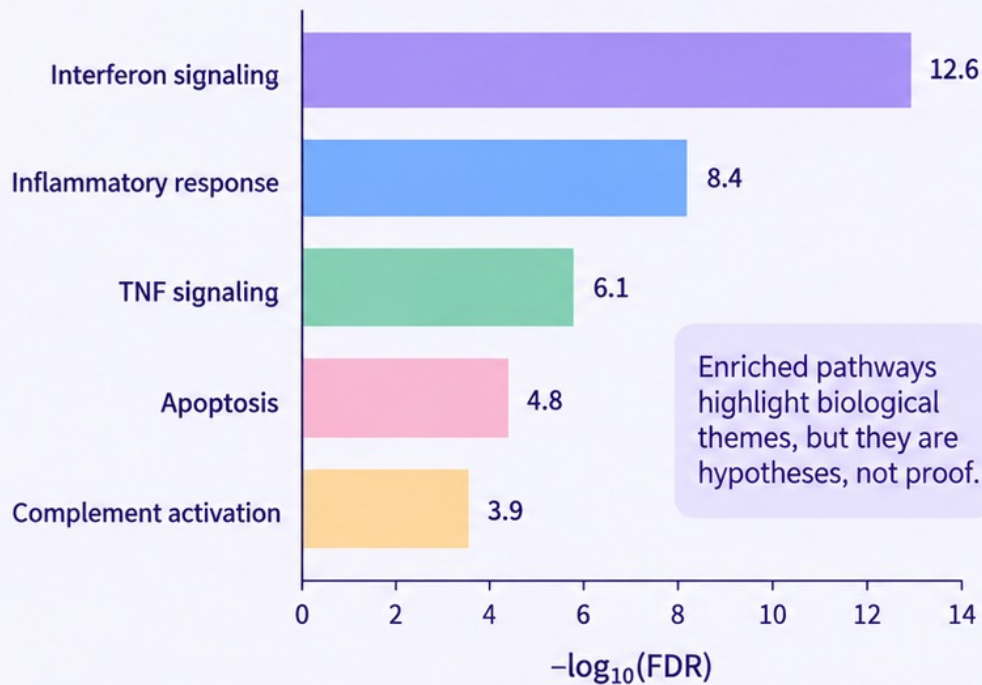
From pathway result to a testable hypothesis

Week 5 | Slide 37

Pathway analysis is a starting point; interpretation should end with an experiment you can design.

Top enriched pathways

Example results from differential expression analysis



1



Enriched pathway

Identify significantly enriched pathways from your gene list (e.g., Interferon signaling).

Example:

Interferon signaling is enriched ($\text{FDR} = 1.2 \times 10^{-6}$).

2



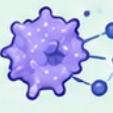
Leading-edge genes

Examine the core genes driving the enrichment (leading-edge subset).

Example:

ISG15, IFIT1, MX1, OAS1 are among the leading-edge genes.

3



Biological interpretation

Connect the pathway and genes to a biological mechanism in your system.

Example:

Suggests an activated innate immune response (e.g., viral response or inflammatory signaling).

4



Testable hypothesis

Convert the interpretation into a specific, falsifiable hypothesis that you can test experimentally.

Example:

The treatment induces an interferon response in these cells, leading to increased antiviral gene expression.

Follow-up experiments

Use targeted experiments to test your hypothesis and validate the pathway's role.



qPCR
validation



Perturbation
or drug treatment



Measure
downstream
phenotype



Enrichment results should generate **hypotheses**, not final conclusions.

Beyond mRNA: the same statistics, different upstream biology

Week 5 | Slide 38

Different RNA classes require different library strategies, annotation, and upstream analysis.

mRNA protein-coding



-  protein-coding genes
-  poly(A) or total RNA
-  DESeq2/edgeR

lncRNA long non-coding



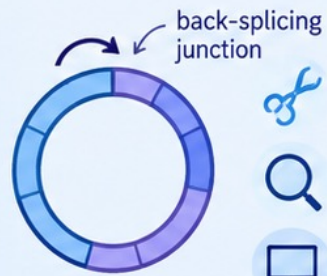
-  > 200 nt
-  regulatory / chromatin roles
-  genome-guided annotation

miRNA microRNA



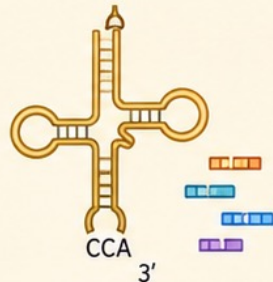
-  ~22 nt
-  small RNA-seq
-  target repression

circRNA circular RNA



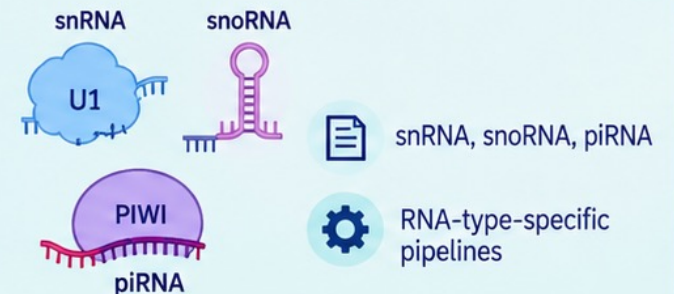
-  back-splicing
-  specialized junction detection
-  CIRIquant / CIRCexplorer

tRNA / tRF transfer RNA



-  translation and fragments
-  specialized library / mapping
-  stress response

other ncRNAs diverse classes



-  snRNA, snoRNA, piRNA
-  RNA-type-specific pipelines

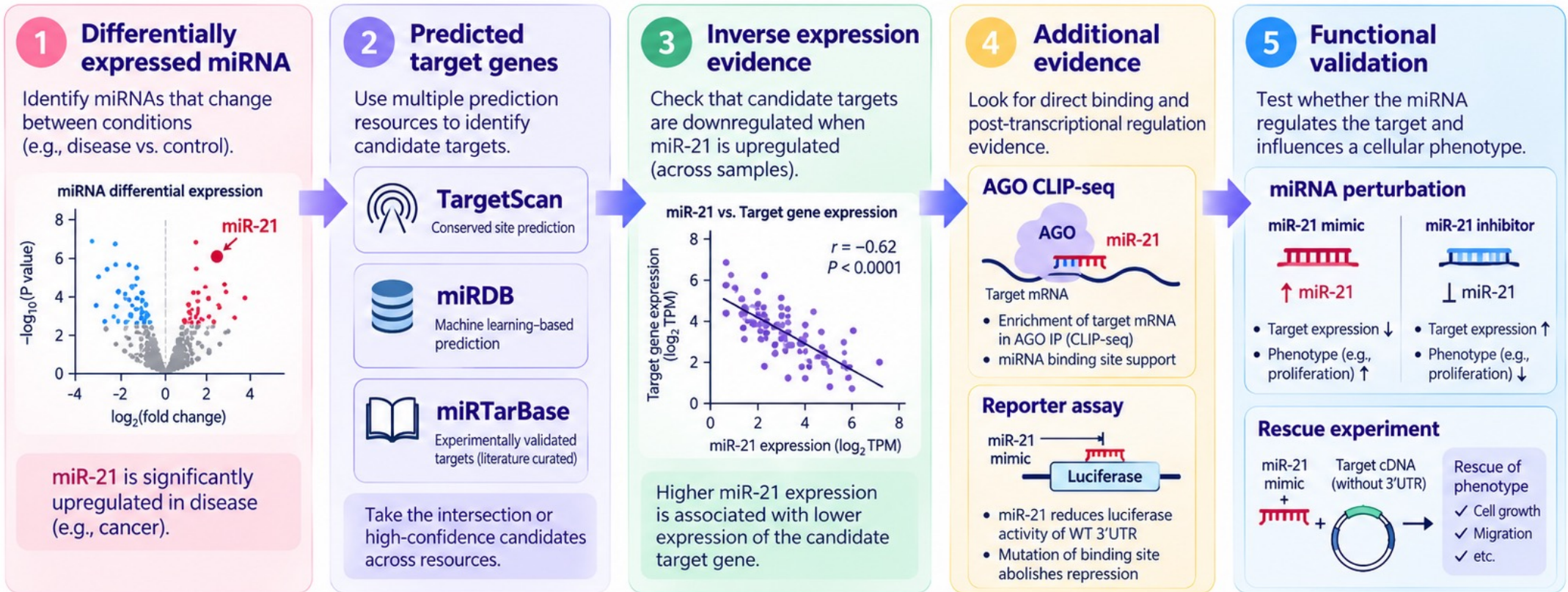


Downstream statistics may be similar, but **upstream detection and quantification** depend on RNA class.

Example: miRNA -> candidate target -> evidence ladder

Week 5 | Slide 39

A regulatory claim should be supported by multiple lines of evidence, not prediction alone.



Key message:

Combine computational prediction, expression evidence, and functional validation before claiming a miRNA-target relationship.

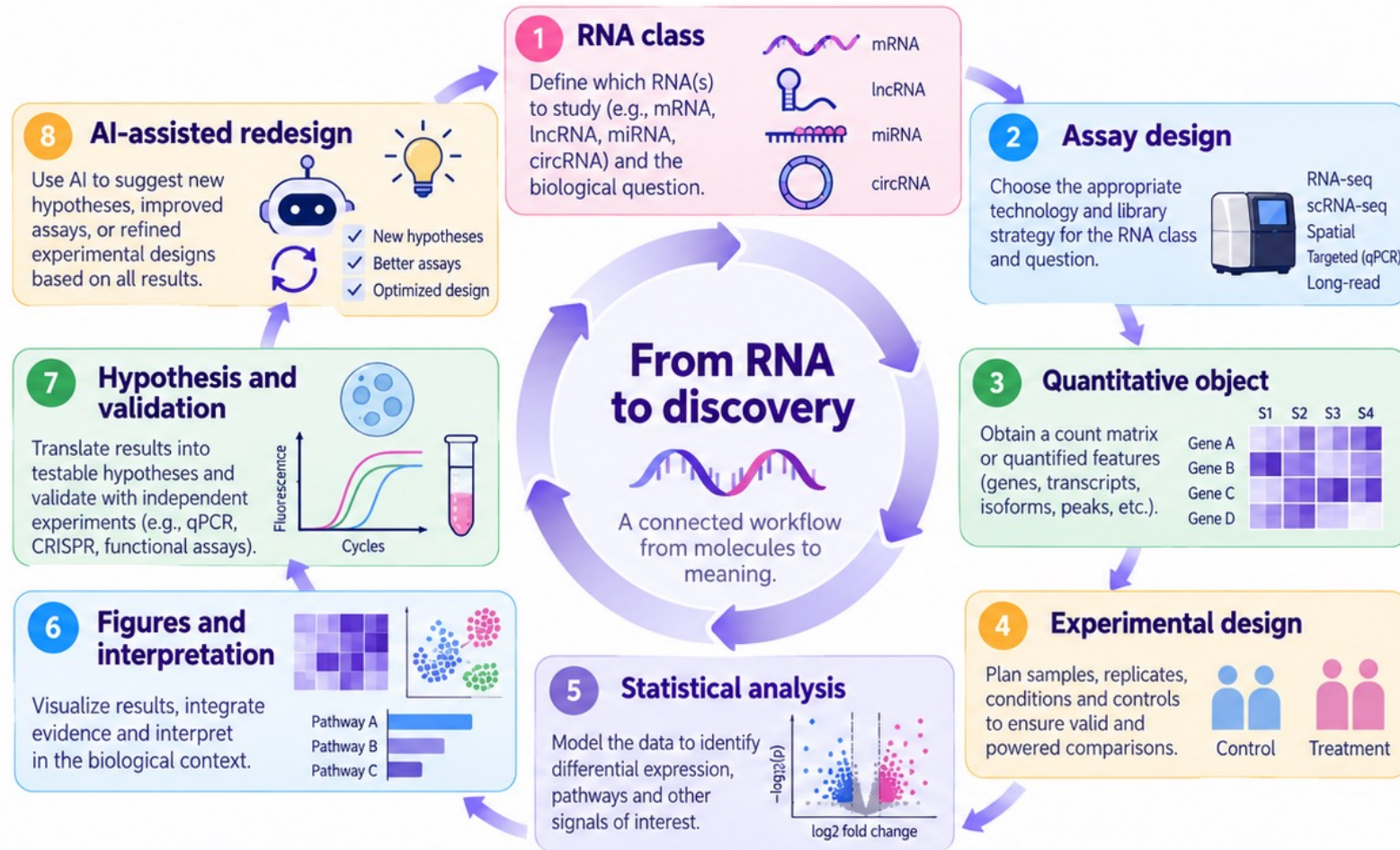


Prediction is only the **first rung** of the evidence ladder.

The complete transcriptomics reasoning loop

From RNA class to validation, every step should connect back to the biological question.

Week 5 | Slide 40



Take-home message



Different RNA classes require different assays.



Statistics turn counts into evidence, not proof.



Interpretation leads to testable hypotheses.



AI can help at every step, but scientific reasoning stays human.

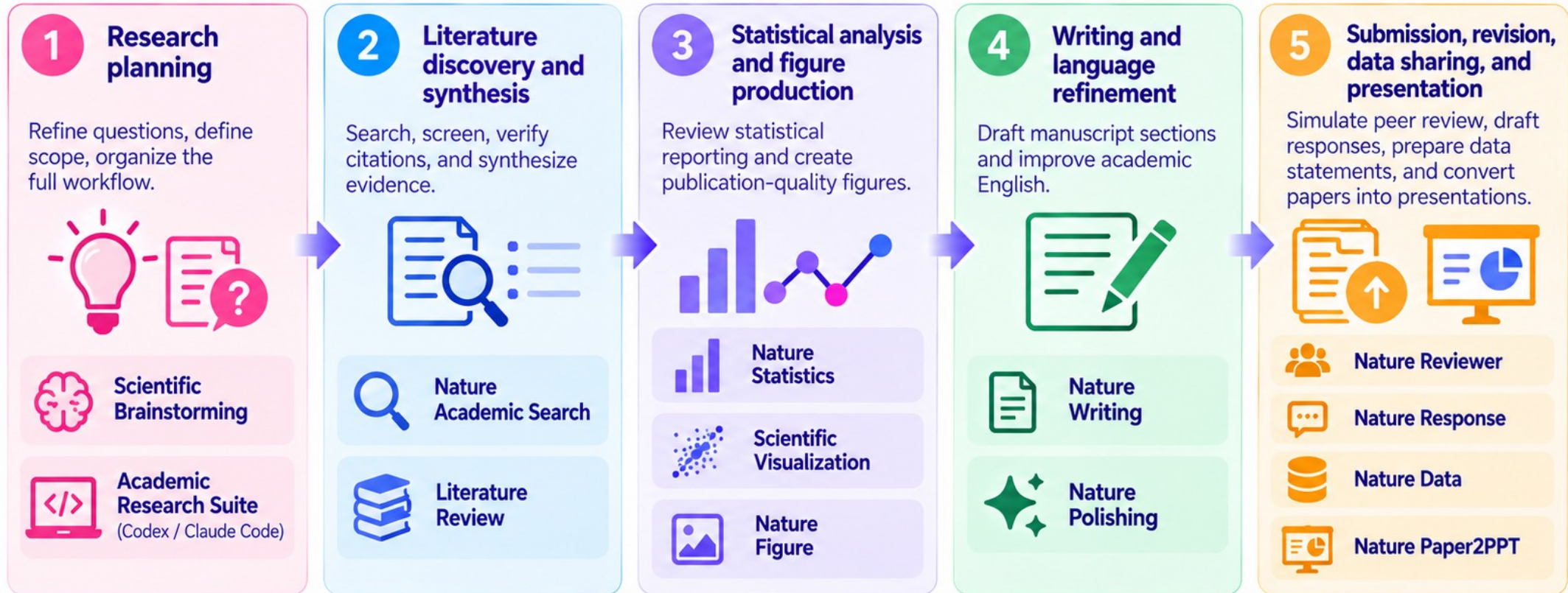


Follow the evidence chain from molecules to hypothesis to validation.

GitHub research AI skills: from idea to publication

Week 5 | Slide 43

Regroup 13 reusable skills into 5 major blocks for a clear, teachable research workflow.



Source structure: 10 research stages · 13 GitHub skills



Human-in-the-loop: AI accelerates, scientists verify.



From idea to impact — use AI skills, but **scientific judgment and verification** remain essential.

科研 AI Skills 工具索引

10 个科研环节 · 13 个技能 · GitHub 完整网址

环节	名称	GitHub 链接	对应用途
1. 选题	Scientific Brainstorming	https://github.com/K-Dense-AI/scientific-agent-skills/tree/main/skills/scientific-brainstorming	细化研究问题，提出可检验的科学假设
2. 文献检索	Nature Academic Search	https://github.com/Yuan1z0825/nature-skills/tree/main/skills/nature-academic-search	多源文献检索、引用核验与文献管理
3. 写综述	Literature Review	https://github.com/K-Dense-AI/scientific-agent-skills/tree/main/skills/literature-review	筛选文献、归纳主题并综合证据
4. 全流程统筹	Academic Research Suite Codex 版	https://github.com/lmbad0202/academic-research-skills-codex	统筹选题、写作、评审和修订流程
4. 全流程统筹	Academic Research Suite Claude Code 版	https://github.com/lmbad0202/academic-research-skills	统筹选题、写作、评审和修订流程
5. 统计审查	Nature Statistics	https://github.com/Yuan1z0825/nature-skills/tree/main/skills/nature-statistics	审查重复数、P 值、多重比较与效应量
6. 科研绘图	Scientific Visualization	https://github.com/K-Dense-AI/scientific-agent-skills/tree/main/skills/scientific-visualization	制作科研图表，优化配色和多面板布局
6. 科研绘图	Nature Figure	https://github.com/Yuan1z0825/nature-skills/tree/main/skills/nature-figure	Python / R 投稿绘图与图件排版检查
7. 论文写作	Nature Writing	https://github.com/Yuan1z0825/nature-skills/tree/main/skills/nature-writing	组织论文论证，起草摘要至讨论各章节
8. 语言润色	Nature Polishing	https://github.com/Yuan1z0825/nature-skills/tree/main/skills/nature-polishing	学术润色与翻译，统一术语和表达
9. 投稿准备	Nature Reviewer	https://github.com/Yuan1z0825/nature-skills/tree/main/skills/nature-reviewer	模拟同行评审，识别方法与论证问题
9. 投稿准备	Nature Response	https://github.com/Yuan1z0825/nature-skills/tree/main/skills/nature-response	逐点回复审稿意见，起草返修附信
9. 投稿准备	Nature Data	https://github.com/Yuan1z0825/nature-skills/tree/main/skills/nature-data	撰写数据可用性声明，规划数据仓储
10. 文献汇报	Nature Paper2PPT	https://github.com/Yuan1z0825/nature-skills/tree/main/skills/nature-paper2ppt	将论文整理为中文 PPTX 文献汇报

Block 1 — Research planning

Week 5 | Slide 44

Turn vague ideas into a focused biological question, hypothesis, and workable project plan.

1

From idea to plan

A simple workflow for research planning.



Problem or observation

What do you notice? What is unclear or interesting?



Research question

Turn the idea into a clear, focused question.



Hypothesis

Propose a testable explanation (prediction).



Study scope

Define scope, key variables, methods, and feasibility.

Goal: make the question testable and the workflow manageable.

2

Scientific Brainstorming

Use AI to explore and refine ideas.



- ✓ Refine vague ideas
- ✓ Compare possible directions
- ✓ Generate testable hypotheses
- ✓ Clarify variables and assumptions

Typical output:

question statement + hypothesis list

3

Academic Research Suite (Codex / Claude Code)

Use AI to organize and execute the plan.



- ✓ Coordinate the full workflow
- ✓ Track literature, writing, and revision
- ✓ Break large tasks into steps
- ✓ Serve as a cross-stage research copilot

Typical output:

scoped workflow plan + task checklist



Human checkpoints

Pause and verify before moving forward.

1

Is the question specific?

2

Is the unit of analysis clear?

3

Is the hypothesis biologically meaningful?



Good AI planning does not replace scientific thinking — it makes **scientific thinking more explicit**.

Block 2 — Literature discovery and synthesis

Week 5 | Slide 45

Search efficiently, verify evidence, and turn many papers into a structured knowledge base.

1

A 4-step literature workflow

From search to a structured knowledge base.

Search



Search across literature databases and sources.

Screen



Filter for relevance and remove irrelevant papers.

Verify citations



Check citation accuracy and trace key references.

Synthesize evidence



Extract insights, compare findings, and build a coherent summary.

Key outputs



Evidence table

Key papers and findings



Gap map

What is known and unknown



Key references

High-confidence sources

2

Nature Academic Search

AI-powered search and validation across the scientific literature.



Search multiple literature sources

Access journals, preprints, and other sources in one place.



Identify highly relevant papers

Get focused results using natural language queries.



Check reference accuracy

Verify citations and trace original sources.



Support citation management

Export and organize references for your workflow.

Best for: finding and validating evidence quickly.

3

Literature Review

Help organize, compare, and write from multiple papers.



Group papers by theme

Cluster related work and identify major topics.



Compare methods and findings

Highlight similarities, differences, and trends.



Summarize consensus and controversy

Identify what is well supported and where there is disagreement.



Draft a cited review document

Generate structured text with in-text citations.

Best for: building a narrative review from evidence.



Human checkpoints

Ask and verify:

1

Are sources relevant?

2

Are citations real and correct?

3

Did we separate evidence from speculation?

4

What knowledge gap remains?



AI can speed up literature work, but **evidence quality and interpretation** still require human judgment.

Block 3 — Statistical analysis and figure production

Week 5 | Slide 46

Audit statistical rigor and convert results into clear, publication-quality scientific figures.



1 Nature Statistics

Functions:

- ✓ Check sample unit and replicates
- ✓ Review P values and multiple comparisons
- ✓ Inspect effect size and confidence intervals
- ✓ Audit statistical reporting



Apply rigorous statistical checks to strengthen the credibility of results.

2 Scientific Visualization

Functions:

- ✓ Choose appropriate chart types
- ✓ Improve color, labels, and annotations
- ✓ Organize multi-panel layouts
- ✓ Make figures easier to interpret



Transform data into clear, accurate, and insightful visualizations.

3 Nature Figure

Functions:

- ✓ Create publication-grade figures in R or Python
- ✓ Inspect panel alignment and spacing
- ✓ Check text collisions and readability
- ✓ Prepare clean final figure panels



Produce polished, publication-quality figures ready for submission.



Typical outputs



Statistical audit checklist



Improved plots



Submission-ready multi-panel figures



Human checkpoints

- ✓ Do the statistics match the design?
- ✓ Do the figures faithfully represent the data?
- ✓ Can the audience understand the message quickly?



Better figures come from better statistical thinking — not decoration alone.

Block 4 — Writing and language refinement

Week 5 | Slide 47

Turn data and figures into a clear scientific narrative, then polish the language for accuracy and readability.



Data and figures

Summarize key results from analyses.



Draft sections

Turn results into a structured manuscript.



Improve logic

Strengthen flow, clarity, and arguments.



Polish language

Refine wording for accuracy and readability.

1

Nature Writing

- ✓ Draft abstract, introduction, results, and discussion
- ✓ Organize arguments around the data
- ✓ Build transitions between sections
- ✓ Convert notes into manuscript text



Typical output:

structured manuscript draft

2

Nature Polishing

- ✓ Improve academic English
- ✓ Reduce redundancy
- ✓ Check terminology and units
- ✓ Make tone concise and consistent



Typical output:

polished, submission-ready language

3

Useful deliverables



Results paragraph



Figure legend



Response-ready text

4

Human checkpoints



- ✓ Are claims supported by data?
- ✓ Is the logic coherent?
- ✓ Are terms and units consistent?
- ✓ Would reviewers understand the argument?



Strong scientific writing connects **evidence**, **logic**, and **language** into one coherent story.

Block 5 – Submission, revision, data sharing, and presentation

Week 5 | Slide 48

Prepare for peer review, respond effectively, manage data responsibly, and communicate the story clearly.



Review



Respond



Share data



Present

1 Nature Reviewer

- ✓ Simulate peer review before submission
- ✓ Identify logic and method gaps
- ✓ Check consistency across the manuscript
- ✓ Highlight likely reviewer concerns



Example output:

Reviewer-style report with key comments

2 Nature Response

- ✓ Parse reviewer comments
- ✓ Draft point-by-point replies
- ✓ Match responses to revised text
- ✓ Check consistency between rebuttal and manuscript



Example output:

Response letter with tracked changes

3 Nature Data

- ✓ Draft data availability statements
- ✓ Plan repositories and access
- ✓ Check FAIR data management requirements
- ✓ Support transparent sharing



Example output:

Data availability statement and repository plan

4 Nature Paper2PPT

- ✓ Turn a paper into a presentation deck
- ✓ Select key figures and methods
- ✓ Organize story flow for teaching or journal club
- ✓ Summarize conclusions clearly



Example output:

Slide deck with key figures and take-home messages

Typical outputs



Mock review



Response letter



Data statement



Presentation slides

Human checkpoints



Did we answer the reviewers directly?



Are shared data complete and compliant?



Do slides faithfully reflect the paper?

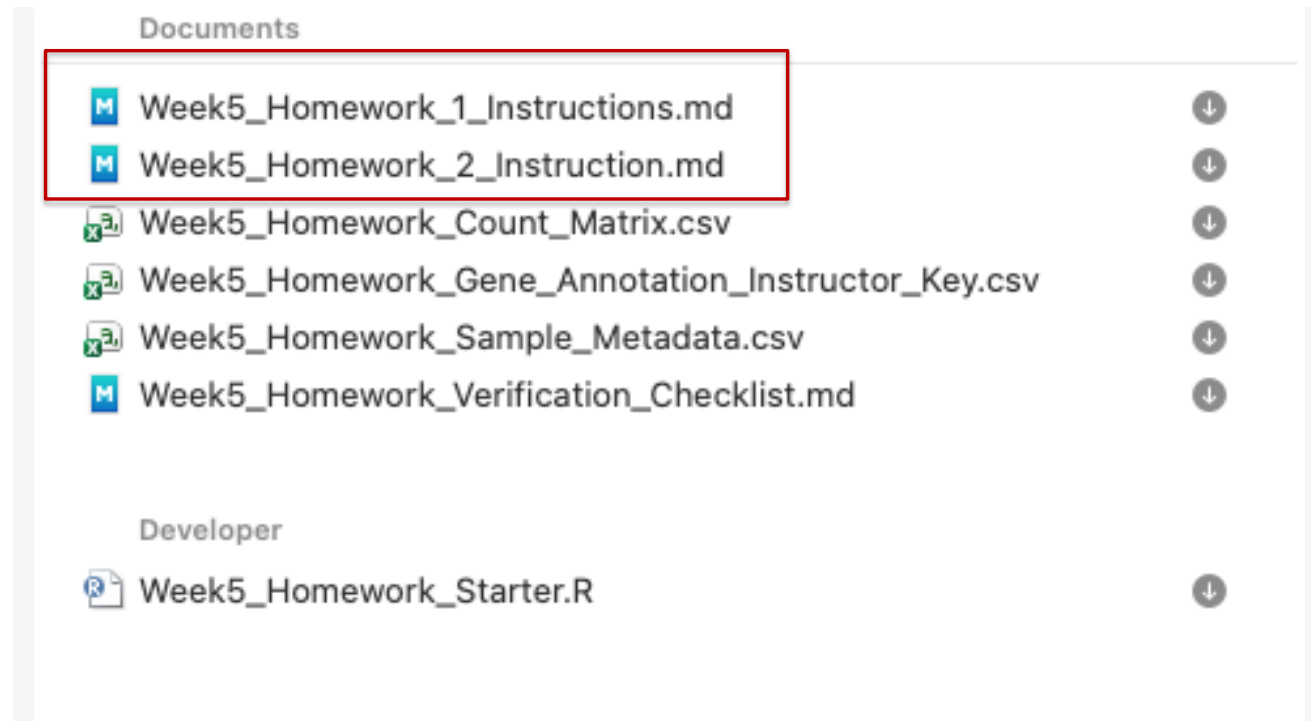


What is the final submission risk?



The final stage is not only submission — it is **verification, transparency, and communication.**

Homework for week 5, **Submission deadline 23:59 on Sunday Night**



We will have quiz next week.

Do not forget to review the class material.

5 multiple choice questions

1 min of each question for total 5 mins